Checkers and Dice

How to Play

The rules of backgammon, game plans and how to use Checkers and Dice: from scratch, step by step

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Checkers and Dice — How to Play

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About this book

This book explains backgammon from scratch, step by step, to someone who has never played it. It has six chapters: an introduction to the game, the board and setup, the rules, playing better, using Checkers and Dice, and reference. If you read the chapters in order, by the end you'll be able to play your first game by the rules; readers who already know backgammon can go straight to Chapter 3 or 4.

How should you read the images? The images come from Checkers and Dice itself and show the board as it appears in the game. You are at the bottom; the cream checkers are yours and the dark brown ones are your opponent's. Your home board is at the bottom right, and the checkers you bear off go to the green tray on the right of the board. Point numbers, arrows and colored frames were added to the images afterward; the board in the app has no numbers by default. The numbers are always given from your side: 1 is the rightmost point of your home board, and 24 is the rightmost point of your opponent's home board. The arrows show where a checker goes, and the small number on an arrow shows which die it was played with; the dice on the board show the numbers rolled that turn.

Two rule sets. Checkers and Dice supports both the international rules (with the doubling cube) and the common Turkish tavla rules. The basic rules are the same in both; where they differ, the relevant sections say so, and Chapter 2 includes a comparison table (Table 2.1).



CHAPTER 1

Getting started

This chapter is for readers who have never played backgammon: a seven-item quick start, a short history of the game and its relatives, the object of the game and the equipment you need.

1.1 Quick start

If you want to understand the game right away without going into details, these seven items are all you need. Each one is covered in full in the chapters that follow.

1. **Goal:** Move your 15 checkers around the board into your home board at the bottom right, then take them off the board. The first to finish wins.
2. **Turn:** On each turn you roll two dice. Each die moves one checker that many points. If both dice show the same number (doubles), you make four moves.
3. **Direction:** Your checkers always move toward lower numbers, toward your home board. There is no going backward.
4. **Where you can land:** On an empty point, on your own checkers or on a single opposing checker. You can't land on a point held by two or more opposing checkers (a made point).
5. **Hitting:** If you land on an opposing checker that stands alone, you hit it, and it goes to the bar in the middle. If your checker is hit, your first job when

your turn comes is to enter it in your opponent's home board; while a checker of yours is waiting on the bar, no other checker can move.

6. **Obligation:** You must play any die you can play; if the two won't both fit, you play the higher one.
7. **Bearing off:** Once all your checkers are in your home board, you start taking checkers off from the point that matches the number rolled. If you finish before your opponent has borne off a single checker, you win a gammon and score 2 points.

1.2 A short history of backgammon

Backgammon belongs to one of the oldest families of games still played today. Games played with dice and checkers, in which the checkers race to the end of a track, have been known for about 5,000 years; today's rules took shape slowly over the centuries, on a journey that runs from Mesopotamia to Rome and from Iran to the coffeehouses of the Ottoman Empire.

Early traces. Traces of race games driven by dice go back to the oldest known settlements. Excavations at Shahr-e Sukhteh in southeastern Iran brought to light a game set about five thousand years old: a board with its track drawn as the coils of a snake, a pair of dice and pieces carved from agate and turquoise, reported to number sixty. How the find was played and how it relates to backgammon are debated. In Mesopotamia, a race-game board of similar age was found in the royal tombs of the city of Ur; because its rules could be partly deciphered from a cuneiform tablet, the Royal Game of Ur is considered the oldest board game whose rules are known. These games are distant relatives of backgammon, not its direct ancestors.

Persian nard. The closest known ancestor of today's backgammon is nard (in Turkish, nerd), played in Sasanian Iran; its origin is placed between the 3rd and 6th centuries AD. According to the widely accepted etymology, even its name pays tribute to a ruler: the word nard is thought to be a shortened form of nēwardaxšīr, meaning "brave Ardashir," a reference to Ardashir I, the founder of the dynasty. The best-loved story credits the invention to Bozorgmehr, the wise vizier of Khosrow I: when chess arrived from India, the vizier figured it out, then designed nard in return and sent it to India, where the Indian sages could not figure it out. The Middle Persian treatise that is the source of this tale describes the game as a model of the universe: the board is the earth, the thirty checkers are the days and nights of the month, the two dice are day and night,

and the opposite faces of the dice, which add up to seven, are the planets. Ferdowsi's Shahnameh tells the same story.

Tabula in Rome and Byzantium. The Romans had a race game called the "game of twelve lines" (ludus duodecim scriptorum), played on three rows of points. Over time the middle row disappeared, leaving a board of 24 points, 12 on each side, and the game came to be called tabula. Each player had 15 checkers. Emperor Claudius is said to have been fond of the game and even to have written a book about it. Today we know its rules largely from a poem describing a game that the Byzantine Emperor Zeno (476–491) lost because of the dice. The main differences between tabula and today's backgammon are that it was played with three dice and had no bar for hit checkers to wait on. The Turkish word "tavla" comes from the same root: Latin tabula (board, table) became "tavli" in Byzantine Greek and passed from there into Turkish.

Nard in the Islamic world. Nard spread from Iran to the Arab world under the same name and for centuries was the game of coffeehouses and caravanserais. Ottoman Turkish, too, used the word "nerd" for backgammon. The liveliest trace of this shared past is in the names of dice rolls: mixed Persian-Turkish names such as "düşeş," "şeşbeş" and "hepyek" are said the same way from the Balkans to the Arab world; in Lebanon and Israel the game itself is even called "şeş beş." The mother-of-pearl inlaid backgammon sets of Damascus have been famous throughout the region since the 19th century.

Europe and "backgammon." Games of the backgammon family were played all over Europe throughout the Middle Ages; the English called them collectively "tables." The game was banned at some times and loved at royal courts at others; the 1283 book of games of King Alfonso X of Castile records the rules of several variants, with illustrations. The name "backgammon" appeared in England in the mid-17th century; its earliest known record is a letter dated 1647. Details such as a hit checker waiting on the bar in the middle and today's scoring of gammons and backgammons had become established by 1850 at the latest, according to the rule books of the period.

The Ottomans and the coffeehouse. Backgammon took root in everyday Turkish life along with the coffeehouse. Istanbul's first coffeehouses opened in Tahtakale in the 1550s and soon spread to every neighborhood. Peçevi, a historian of the period, writes that among the people who gathered in these new places, some read books while others played backgammon or chess; Evliya Çelebi counts hundreds of coffeehouses in 17th-century Istanbul. Across the vast lands under Ottoman rule, backgammon became an inseparable part of

this coffeehouse culture, and it stayed in place after the empire withdrew. Along with the Byzantine, Persian and Arab game traditions, this coffeehouse heritage is an important reason why backgammon is considered almost a national game today in countries such as Turkey, Greece, Cyprus, Egypt and Lebanon.

The modern era. The doubling cube, which appeared in New York gaming clubs in the 1920s, took the game to a whole new level by adding probability and risk management. Backgammon came back into fashion in the 1960s; the first world championship was held in Las Vegas in 1967, and since 1979 the championship has taken place in Monte Carlo. Computers entered this story early: in 1979 a program called BKG 9.8 beat the reigning world champion, Luigi Villa, in a match, although even the program's creator admitted that the dice had favored it. In the 1990s TD-Gammon, which learned by playing against itself, reached the level of master players; today every stage of the game, from opening moves to cube decisions, has been rewritten by the analyses of such programs. The World Backgammon Federation, founded in 2018, is one of the main organizations working to harmonize international tournament rules; in Turkey, too, there are associations that run tournaments with the doubling cube. Today backgammon reaches millions of players through online platforms and mobile apps.

TABLE 1.1 A timeline of backgammon

Period	Event
~3000 BC	A game board with two dice and 60 pieces at Shahr-e Sukhteh
~2600 BC	Royal Game of Ur, Mesopotamia
3rd–6th c. AD	Nard in Sasanian Iran; the legend of Bozorgmehr
AD 476–491	Reign of Zeno; tabula in Byzantium
1283	Backgammon variants in Alfonso X's book of games
1550s	First coffeehouses in Istanbul; backgammon becomes a coffeehouse game
1647	First known use of the name "backgammon"
By 1850	The bar rule and today's scoring established by this date at the latest
1920s	The doubling cube in New York

Period	Event
1967	First world championship, Las Vegas
1979	BKG 9.8 beats the world champion
1990s	TD-Gammon; AI reaches master level
2018	World Backgammon Federation founded; starts harmonizing tournament rules

Note: The story that Bozorgmehr invented the game and the reading of the board as a symbol of the universe come from Sasanian-era texts; historians read them as legends. What is certain is that the game took shape in Iran in this period and spread from there.

1.3 Other backgammon variants

This book covers classic backgammon, the form most widely played in the world today. But a backgammon board is not the board of just one game: from Turkey to the Caucasus and from Greece to Syria, dozens of different games are played on the same 24 points. They fall into three families:

- **Hitting games:** A lone opposing checker is hit and sent to the bar. Classic backgammon and Greek portes belong to this family.
- **Pinning games:** Instead of hitting, you land on top of a lone opposing checker and pin it in place. Mahbusa, tapa and plakoto belong to this family.
- **Races without hitting:** There is no hitting and no pinning; a single checker holds a point, and both sides usually travel in the same direction. Gülbahar, mültezim, fevga and long nardy belong to this family.

Here are the main ones you may come across at the table:

- **Mahbusa ("prison backgammon"):** Known as tapa in the Balkans and plakoto in Greece. If you place a checker on top of a lone opposing checker, you pin it; the checker underneath can't move until you move off it. A checker that is pinning another can't itself be pinned. In the usual setup, all 15 checkers of each player start from the opponent's 1-point. The game is slow and positional: a single checker pinned early can decide the whole game. There is no doubling cube and no backgammon.
- **Gülbahar:** Its international name is gul bara. Even a single checker on a point keeps the opponent from landing there. Both sides' checkers start

from diagonally opposite corners of the board and travel in the same direction. Its most interesting rule concerns doubles: after the first three turns, a player who rolls doubles plays them and then plays all the doubles that follow, in order, up to 6-6; if a double can't be played in full, the rest of the turn is forfeited. A single double can turn the game around in an instant.

- **Mültezim:** The Turkish name for the game known in Greece as fevga. The checkers start from diagonally opposite corners and travel in the same direction; there is no hitting, and a single checker holds a point. It has two special rules: you can't move any other checker until your first checker has crossed into your opponent's starting quadrant, and you can't block all six points of your own starting quadrant at once. Cutting your opponent off completely, so that they have no checker left to move, is also forbidden.
- **Long nardy:** The game of Russia, the Caucasus and Azerbaijan. Each player's 15 checkers start stacked on a single point, the "head"; only one checker may leave the head each turn, but if the first roll is 6-6, 4-4 or 3-3, two checkers leave it. There is no hitting, a single checker holds a point, and building a six-point prime in front of all the opponent's checkers is forbidden. Here too, finishing before the opponent has borne off a single checker counts as a gammon.
- **Greek backgammon (tavli):** Not a single game but a match in which three games are played in turn: portes, plakoto and fevga alternate until one player reaches the target, usually 5 or 7 points. **Portes** is almost identical to classic backgammon; the starting player rolls the dice again, most rule collections forbid hit-and-run in the home board, and there is no backgammon and no doubling cube. **Plakoto** is the Greek form of mahbusa; all the checkers start from a single point, and if the last checker left on the starting point (the "mother") is pinned, the game ends and the pinning side wins a gammon (local rules have exceptions to this). **Fevga** is the counterpart of mültezim.

The detailed rules of these games vary from region to region, and their strategies differ quite a lot. What they share is reading the dice, counting out moves and the goal of bringing the checkers home and bearing them off; details such as the bearing-off requirements can vary from game to game. Once you have learned classic backgammon well, moving on to them is still easy.

TABLE 1.2 The main backgammon variants

Game	Where it is common	Hitting	Direction of movement	Starting position
Classic backgammon	Turkey, the Western world, the Middle East	Yes	Opposite directions	The 2-5-3-5 setup
Mahbusa	Turkey, Syria, Lebanon, the Balkans	Pinning	Opposite directions	15 checkers on the opponent's 1-point
Gülbahar	Turkey, Bulgaria, Azerbaijan	No	Same direction	Diagonally opposite corners
Mültezim / Fevga	Turkey, Greece	No	Same direction	Diagonally opposite corners
Long nardy	Russia, the Caucasus	No	Same direction	Diagonally opposite corners

1.4 The object of the game

Backgammon is a race game for two players. Each player has 15 checkers, and your goal is to move all 15 around the board into your own home board and then bear them off the board one by one. Whoever does this before their opponent wins.

You can think of it as a running race: two teams run on the same track in opposite directions, and the race ends when one team's last runner crosses the finish line. Along the way you can block your opponent's path (by making points) and send a lone runner back to the start (by hitting). Because the players take turns, one of them always finishes first; that's why backgammon never ends in a draw.

The game involves both luck and skill. The dice decide how far you move each turn, but you decide which checkers to play those numbers with. Bad rolls can cost a good player a single game, but over dozens of games it is the choices made, more than the dice, that make the difference.

1.5 Equipment

- **Board:** A box that folds in two on hinges, with 24 long triangles drawn on the inside. Details in section 2.1.
- **Checkers:** Round pieces much like those in the game of checkers. There are 30 in all; each player takes 15, and the two sets are told apart by their color.
- **Dice:** Ordinary game dice with 1 to 6 dots. At home one pair is enough; in official matches each player uses their own pair.
- **Dice cup:** A small cup for shaking and throwing the dice. You don't have to use one, but it makes the roll fair because it leaves no chance to steer the dice by hand.
- **Doubling cube:** A special cube whose numbers double from 2 up to 64. It sits on the table as a reminder of how many points the game is worth. Details in section 3.8.

Traditional Turkish names for dice rolls

At backgammon tables in Turkey, rolls are often called out not by their numbers but by their old names. These names are rooted in Persian and Turkish numerals. You don't need to memorize them; the ones you hear most often are listed below so your ear can get used to them.

TABLE 1.3 Traditional Turkish names for dice rolls

Roll	Name	Roll	Name
1-1	Hep Yek	6-6	Düşeş
2-2	Dubara	6-5	Şeş Beş
3-3	Dü Se	5-4	Cihar ü Penc
4-4	Dört Cihar	4-3	Cihar ü Se
5-5	Dü Beş	2-1	Yek-i Dü

The numbers: yek is 1, dü is 2, se is 3, cihar is 4, penc is 5 and şeş is 6. Most doubles have a name of their own (such as Hep Yek, Dubara and Düşeş). The way they are said varies from region to region; at most tables, some rolls, such as 5-4, are simply called out with Turkish numbers: "beş dört."

CHAPTER SUMMARY

- Backgammon is a race game for two: the first to bring all 15 checkers into their home board and bear them off wins; the game never ends in a draw.
- The dice decide how far you move each turn, and you decide which checker to play; over dozens of games, the choices make the difference.
- The closest known ancestor of today's backgammon is nard, played in Sasanian Iran; the Turkish word "tavla" comes from Latin tabula.
- Other games, such as mahbusa, gülbahar, mültezim and long nardy, are played on the same board.
- What you need: a board with 24 points, 30 checkers and a pair of dice. A dice cup is optional; the doubling cube is used only under international rules.



CHAPTER 2

The board, setup and game screen

In this chapter you'll learn about the quadrants of the board, the starting position of the checkers, how a game begins and how the two rule sets differ; the chapter ends with the Checkers and Dice game screen.

2.1 The backgammon board

When you open the board, you see 24 narrow triangles in all, lined up along its two long sides; each one is called a **point**. The triangles alternate between two colors only to make them easier on the eye; the colors mean nothing in the rules.

Quadrants. Think of the board as a field divided into four quadrants, each with six points. The two quadrants in front of you are your side; the two across from you are your opponent's:

- **Your home board (1–6):** Where the journey ends. Once checkers get here, they are taken off the board. Bottom right in Figure 2.1.
- **Your outer board (7–12):** The last stop before your home board. Bottom left.

- **Opponent's outer board (13–18):** The middle of the route. Top left.
- **Opponent's home board (19–24):** Where the journey begins. Top right.

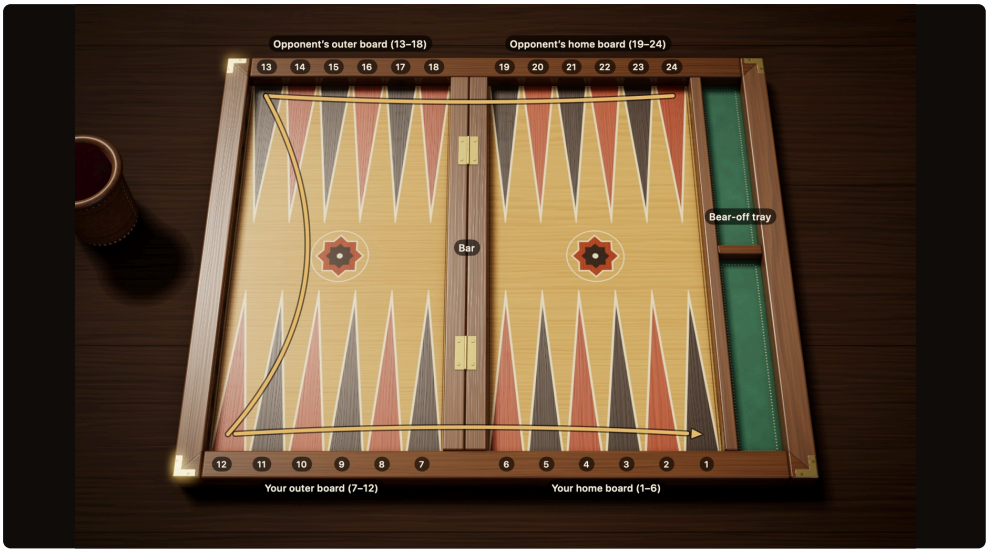


FIGURE 2.1 **The backgammon board: 24 points, four quadrants and the bar.** You are at the bottom. The gold arrow shows the path of your checkers from the 24-point to the 1-point; your opponent travels the same path in the opposite direction.

The bar. The raised strip down the middle of the board is called the bar. It has two jobs: it separates the left and right halves of the board, and it is where hit checkers wait. It doesn't count as a point when you count out a move: a checker moving from the 7-point to the 6-point takes just one step, even though it crosses the bar.

How the numbering works. Each player's numbers start from their own home board. So every triangle has two names: the point you call 1, your opponent calls 24. To avoid confusion, all the numbers in this book are given from your side of the board. Checkers and Dice can also print these numbers on the board's rails: turn on "Point numbers" in the Appearance section of Settings.

Direction of travel. Your checkers always move toward lower numbers. A checker setting out from the 24-point at the top right first moves left along the top edge, turns down at the left edge, then moves right along the bottom edge until it reaches your home board. Your opponent's checkers travel the same

path the other way around, which is why the two sides meet head-on along the way.

Mirrored board. Some players like to set up their home board on the left. The whole board is then mirrored, and the checkers travel clockwise. As far as the rules go, there is no difference at all between the two setups.

2.2 Starting position

Every game begins from the same starting position. Place your 15 checkers on four points, from the farthest to the nearest, like this:

1. **24-point, 2 checkers.** The corner of your opponent's home board. These two are your checkers farthest from home.
2. **13-point, 5 checkers.** The top left corner; the exact middle of the route.
3. **8-point, 3 checkers.** In your own outer board, two steps from your home board.
4. **6-point, 5 checkers.** The first point of your home board, next to the bar.

2 + 5 + 3 + 5 makes 15. Your opponent sets up the same pattern from their own side, so their checkers sit on your points 1, 12, 17 and 19. If you folded the board along the middle, the two sides would match exactly.



FIGURE 2.2 **Starting position: 15 checkers for each player.** Your checkers are cream and your home board is at the bottom right; your opponent's checkers are dark brown, with their home board at the top right. You have 2 checkers on the 24-point, 5 on the 13-point, 3 on the 8-point and 5 on the 6-point.

An easy way to remember it: Memorize “2-5-3-5” and set out the checkers starting from the rearmost point. A third of your checkers (5 checkers) are already in your home board; the rest are at different points along the route. Much of the game is spent bringing the checkers left behind safely home.

2.3 Starting the game

The dice decide who makes the first move. Each player rolls a single die, and whoever rolls the higher number starts. If the numbers are equal, the roll doesn’t count; both players roll again until the numbers differ.

There are two different traditions for the winner’s first move:

- **International rule:** Nobody rolls again. The two dice on the table, yours and your opponent’s, together make your first move. If you rolled 5 and your opponent 3, you play 5-3. Since the opening dice can’t be equal, this first move is never doubles.
- **Common Turkish rule:** The winner picks up both dice and rolls again. This roll can be anything, doubles included.

After that, the players take turns, rolling two dice each turn.

New game. Under international rules, every game starts afresh with an opening roll. In Turkish tavla, the winner of the last game usually opens.

Dice etiquette. There are a few customs when playing at a table. The dice are usually shaken in the cup and rolled into the right half of the board. A die that lands tilted, rests on a checker or jumps off the board can’t be played; the roll is repeated with both dice. Your turn begins the moment your opponent picks up their dice; a roll you make before then doesn’t count. What is described here is traditional table etiquette; tournament rules may differ in these details depending on the use of clocks, dice cups or four dice.

2.4 Turkish and international rules

In both sets, counting out moves, the dice rules and the logic of made points are the same. The differences lie in the start, the scoring and two details of play: in the common form of Turkish tavla, hit-and-run in your own home board is considered forbidden, and so is wasting pips during the bear-off when you could

take a checker off. The table below follows the common rules; there may be small differences from table to table.

In Checkers and Dice you can play with either rule set. Wherever the two sets differ, this book gives both rules; all you need to do is choose which one you'll play before the game begins.

TABLE 2.1 Turkish tavla and international rules

Topic	International (with cube)	Common Turkish tavla
First move	Played with the opening dice; can't be doubles	The higher roller rolls both dice again; can be doubles
Who starts the next game	An opening roll is made in every game	The winner of the previous game starts
Hit-and-run in your own home board	Allowed	Considered forbidden (in Checkers and Dice, set for each new match; allowed by default)
Moving a checker on during the bear-off when you could bear one off ("sallamak")	Allowed	Considered forbidden (in Checkers and Dice, set for each new match; allowed by default)
Backgammon	Yes (× 3)	No; counts as a gammon (× 2)
Doubling cube	Standard; compulsory in tournaments	None at all
Crawford rule	Standard in matches	None

2.5 The Checkers and Dice game screen

When a match starts, areas showing the state of the game appear around the board. Figure 2.3 was captured during your turn; the numbers are explained below it.

1. **Opponent's card:** Shows your opponent (with the difficulty level, as in "Computer · Medium"), their pip count, their time left if the match clock is on, and their score in the match. "Thinking…" appears while the computer is thinking.
2. **Menu:** Pauses the game. The card that opens lets you resume, open the settings or the "How to Play" card, see the move history, your matches and the

match equity table, resign or leave the match. The Esc key does the same.

3. **Match info:** Shows how many points the match is played to (in a session it says "Session") and which game you are in. Under international rules, the cube indicator sits below it. The bottom line names the type of position, such as "Running game," "Blitz," "Back game" or "Holding game" (game plans are in section 4.4). Tap the match info (click it on the Mac) and a one-sentence explanation of the plan appears in the hint line; you can turn the bottom line off with "Show position type" in Settings.
4. **Dice:** Your dice land in the right half of the board, the computer's in the left half.
5. **Bear-off tray:** The green tray on the right of the board. Your borne-off checkers line up in its lower half, your opponent's in its upper half.
6. **Your card:** Shows your pip count, your time left if the match clock is on, your match score and, when it's your turn, the "Your turn" notice.
7. **Hint line:** Briefly tells you what you can do at that moment. If you try an illegal move, this is also where you read why, for example "This checker can't move with these dice."



FIGURE 2.3 **The Checkers and Dice game screen.** The numbered items are explained in the list below.

Optional buttons. If you turn them on in Settings, three more buttons appear at the bottom right: "Hint" marks on the board the move suggested by the Master engine (⌘I); "Undo" takes back the steps you've played this turn (⌘Z);

and “Done,” when auto-confirm is off, ends your turn and passes it to your opponent (Return). Details in section 5.3.

Before you start. In the New Match panel you choose the difficulty level (or the “Adaptive” opponent, which sets its strength to suit you), the rule set (Turkish tavla or international), a match clock if you want one, the match length or, under international rules, a session, and the color of your checkers. These choices apply for the whole match; details in section 5.1.

How does a turn work? When it’s your turn, the dice are rolled for you and pour out of the leather cup onto the board; under international rules, on a turn when you could double, the app waits and you roll with the “Roll Dice” button or the Space bar. You move a checker by dragging and dropping it, or by clicking first the checker and then the point it should go to; the checkers you can move glow, and the points they can reach are marked with rings. Once you play your last step, your turn ends and passes to your opponent. If you’d rather roll every turn yourself, confirm your turn with “Done” or take steps back, you choose these in Settings; details in section 5.3. The app won’t accept an illegal move anyway. If you’d rather roll real dice on the table yourself, turn on “Play with real dice” in Settings and enter the faces (section 5.2).

What does the pip count tell you? The pip count on the cards is how many more points that player must travel to bring all their checkers home and bear them off. The player with the lower count is ahead in the race. Details in section 4.2.

CHAPTER SUMMARY

- The 24 points of the board are divided into four quadrants of six: your home board 1–6, your outer board 7–12, your opponent's outer board 13–18, your opponent's home board 19–24.
- Each player's numbers start from their own home board; all the numbers in this book are given from your side.
- The starting position is "2-5-3-5": 2 checkers on the 24-point, 5 on the 13-point, 3 on the 8-point and 5 on the 6-point.
- Who moves first is decided with one die each. Under the international rule the opening dice are played; under the common Turkish rule the winner rolls both dice again.
- The two rule sets differ in the start, the scoring and two details of play; Table 2.1 sums up the differences.
- In Checkers and Dice, the pip count, match info, menu and hint line appear around the board (Figure 2.3).



CHAPTER 3

The rules of the game

This chapter covers all the rules of the game in order: moving the checkers, doubles, forced moves, hitting and entering from the bar, bearing off, scoring and the doubling cube.

3.1 Moving the checkers

The player whose turn it is rolls two dice and moves checkers forward by the numbers on the dice. This section explains step by step how to play a roll.

Reading the dice

The dice tell you how many points your checkers move. A die showing 4 means one checker moves 4 points. When counting, you don't count the point the checker is standing on; you start from the next one. A checker on the 13-point playing a 4 counts 12, 11, 10, 9 and lands on the 9-point.

Backgammon has no reverse gear: a checker can only move toward lower-numbered points, that is, closer to its home board.

Two dice, two separate moves

Think of the two dice you've rolled as two separate moves you're entitled to. When you roll 5-3, you can use them however you like:

- You can play the 5 with one checker and the 3 with another.
- You can give both to the same checker and move it 8 points in total; you can play either the 3 first or the 5 first.

When you play both dice with one checker, the checker actually makes two separate hops and briefly touches down on the point where the first hop ends. That intermediate point must also be a point the checker can land on. You can't count the total of the two dice in one go and skip the point in between.

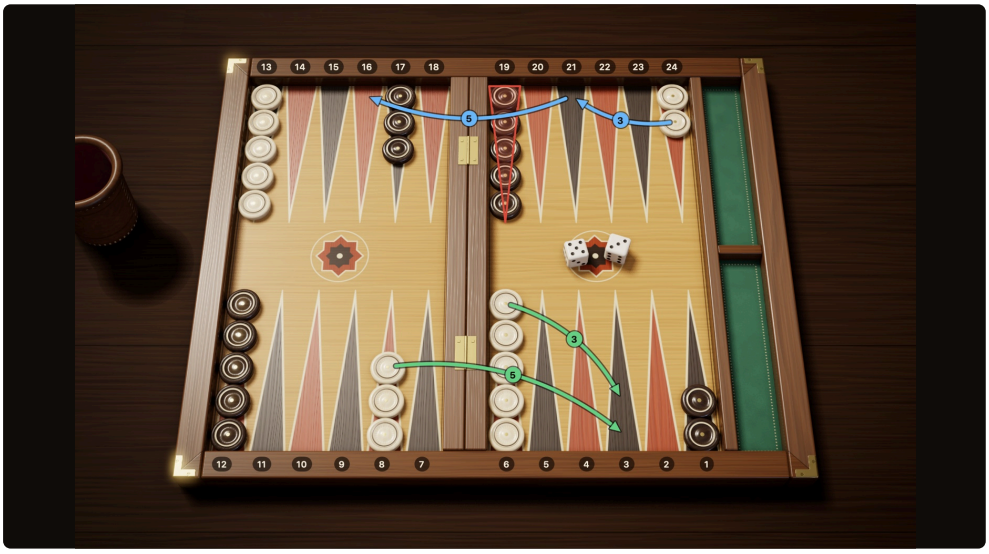


FIGURE 3.1 **Two ways to play a 5-3.** Green arrows, option 1: 8→3 (5) and 6→3 (3); two different checkers make the 3-point. Blue arrows, option 2: 24→21 (3), then 21→16 (5); the same checker moves 8 points in total. That checker can't play the 5 first: the 19-point, framed in red, is a point your opponent has made and can't serve as an intermediate point.

The reverse is also true: you can't split a single die into parts. Sharing a 5 between two different checkers as a 3 and a 2 is against the rules; each die belongs, whole, to a single checker.

Where can a checker land?

When you move a checker, the state of the destination point matters. There are four possibilities:

TABLE 3.1 Can a checker land on the destination point?

What is on the destination point?	Can a checker land there?
No checkers (an empty point)	Yes
One or more of your own checkers	Yes, with no limit on the number
A single opposing checker (a blot)	Yes, and the opposing checker is hit
Two or more opposing checkers (a made point)	No



FIGURE 3.2 **Open points and a made point.** For the checker on the 11-point: the 10-point is empty, so it can land there. The 8-point holds your own checkers; you can add as many as you like. The 7-point holds a single opposing checker (a blot); you can land there, and that checker is hit. The 9-point, framed in red, holds two opposing checkers (a made point): the dashed red arrow shows that you can't land there.

A point where two or more of a player's checkers stand together is called a **made point**. Think of it as a roadblock: your opponent can neither land on it nor touch down on it as the intermediate point of a two-dice move. Making your own points wisely is the most effective way to narrow your opponent's path.

3.2 Doubles

When both dice show the same number (from 1-1 to 6-6), you get a bonus: you use that number not twice but a full four times. A player who rolls 5-5 makes four separate moves of 5 and moves 20 points in total. These rolls are called **doubles**.

You can split the four moves however you like: once each for four different checkers, twice each for two checkers or four times for a single checker. If one checker takes more than one step, every intermediate point it touches down on must be open.

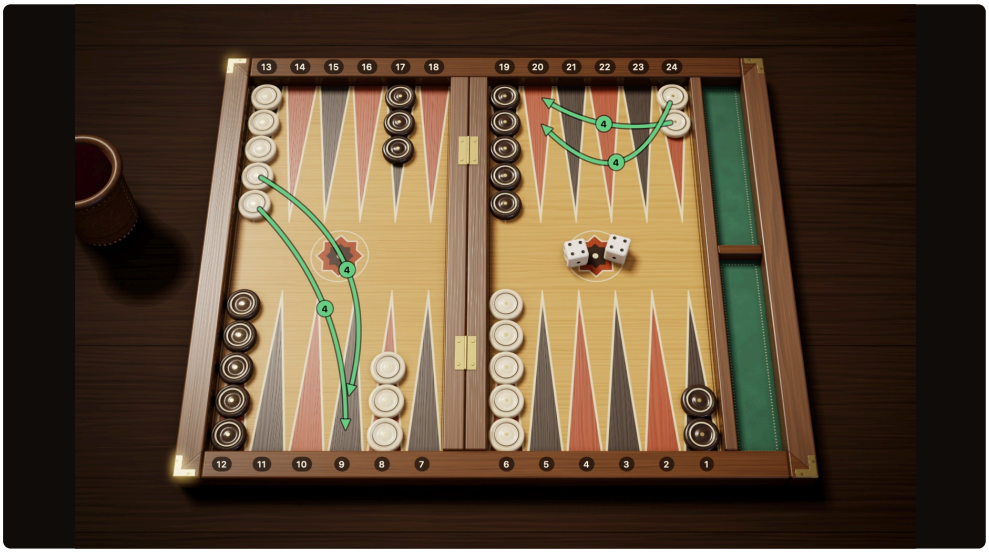


FIGURE 3.3 **Doubles: four moves with 4-4.** Move the two checkers on the 24-point to the 20-point (4 + 4) and bring two checkers down from the 13-point to the 9-point (4 + 4). Each of the four moves is 4 points; you can split them among whichever checkers you like.

3.3 Forced moves

Once you've rolled, you have no right to say "I'll skip this turn." If the dice allow a move, you must make it. These questions help you work out which dice you must use, and how many:

1. **Is there a move that uses both dice?** If so, you must make it. If the first move that comes to mind leaves one die unused, look for another way that

fits both dice.

2. **Can only one die be played?** Then you use the one that can be played, and the other is forfeited.
3. **Can each die be played on its own, but not both together?** Then the choice isn't yours: you use the higher die. Figure 3.4 shows this.
4. **Can neither die be played?** Your turn is lost, and it's your opponent's turn to roll.
5. **Did you roll doubles?** If not all four moves fit, you must make as many moves as can legally be played; if the order of the moves affects that number, you must choose the order that lets you play more. The rest are forfeited.



FIGURE 3.4 **The higher-die rule: 6-5.** The checker on the 14-point can go to the 8-point with the 6 (green arrow) or to the 9-point with the 5 (dashed arrow), but it can't play both: either way, the second step lands on the 3-point (red frame), a point your opponent has made. Your other checkers, deep in your home board, can't play either die. When the dice can be played separately but not together, the higher die is played: 14→8.

As a result of these rules, sometimes there is only one move you can make; this is called a **forced move**. In digital backgammon the app itself enforces these rules and won't allow an illegal move; on a real board, it's up to you to pay attention.

3.4 Blots and hitting

What makes backgammon exciting is hitting. First, you need to tell two situations apart:

- **Blot:** A checker left alone on a point. It is unprotected; if your opponent lands on it, it is knocked off its point.
- **Made point:** A point shared by at least two checkers of the same color. Checkers standing there can't be touched.

When you bring a checker down on the point where your opponent's blot stands, you have **hit** that blot. The hit checker is lifted off its point and placed on the bar in the middle. For your opponent this is a serious setback: the checker has to start its journey all over again, and until it returns to the board, your opponent's hands are tied.

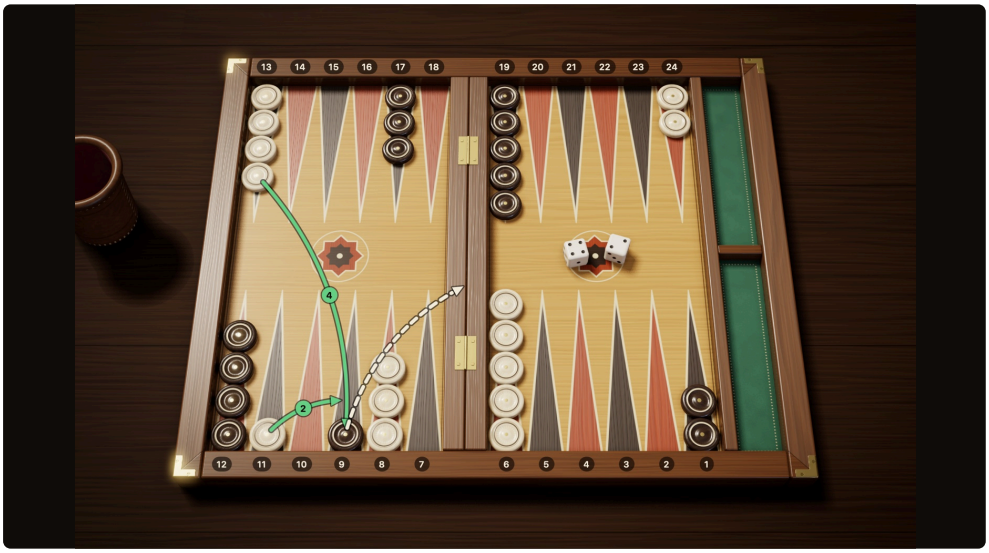


FIGURE 3.5 **Hitting: a blot is hit with 13→9.** With the 4, 13→9: you land on your opponent's single checker and hit it; the hit checker goes to the bar (dashed arrow). With the 2, 11→9: you bring a second checker there too and make the 9-point; now you can't be hit.

What you need to know:

1. Hitting is not compulsory. Even when you could hit a blot, you can choose to make a different move.

2. After hitting, you can carry on with the same checker. For example, with 4-2 you can move a checker 4 points to hit, then move the same checker on another 2 points. This is called "hit-and-run." Under international rules it is allowed anywhere on the board; in the common form of Turkish tavla and in Greek portes, however, running on with the same checker after hitting in your own home board is forbidden. In Checkers and Dice, when you play by Turkish rules, you can turn this restriction on with the "No hit-and-run" option in the New Match panel; it is off by default.
3. You can hit more than one blot in the same move; for example, a checker can hit one blot at the intermediate point of the two dice and another where it lands.
4. The same rules apply to you. A checker you leave as a blot is hit if your opponent's roll reaches it. So whenever possible, try to keep your checkers in pairs.

3.5 Entering from the bar

When one of your checkers has been hit, there is only one item on your agenda when your turn comes: getting that checker back onto the board. As long as you have a checker waiting on the bar, your other checkers are frozen.

Where does it enter? A checker on the bar starts its journey over, that is, from your opponent's home board. The die decides which point it enters on. Count your opponent's home board from 1 to 6, from the edge toward the bar: with a 1 you enter on the point in the corner of the board, with a 6 on the point next to the bar. In your numbering:

TABLE 3.2 Where a checker on the bar enters, by die

Die	Entry point
⚀ 1	24
⚁ 2	23
⚂ 3	22
⚃ 4	21
⚄ 5	20

Die	Entry point
⚅ 6	19

The point shown by the die may be empty, it may hold your own checkers, or a single opposing checker may be standing there. In all three cases you enter; in the last case you hit that checker as you come in. Only a point your opponent has made is closed to you.

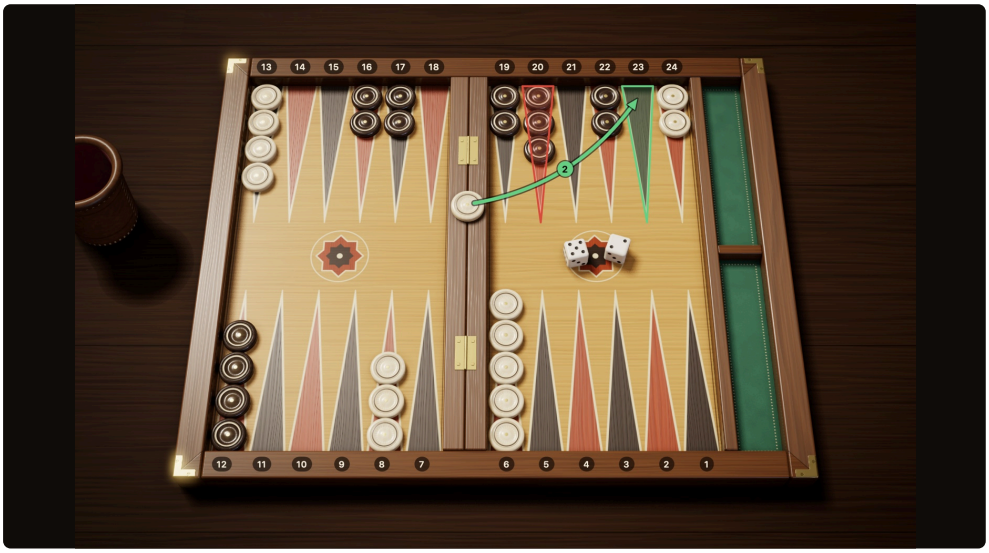


FIGURE 3.6 **Entering from the bar: 5-2.** Your checker on the bar enters in your opponent's home board: with a 1 on the 24-point, with a 2 on the 23-point … with a 6 on the 19-point. You can't enter with the 5: your opponent has made the 20-point (red frame). With the 2 you enter on the empty 23-point (green frame). Once you've entered, you play the 5 with another checker or with the one that just came in.

Entering, step by step:

1. Roll the dice and see which point in your opponent's home board each die corresponds to.
2. If at least one of those points is open to you, place your checker there.
3. Entering uses one die. The other die is free; you play it with any checker you like, including the one that has just entered.
4. If you have two checkers on the bar, both dice go to entering them. If only one can enter, you bring that one in; since the second checker is still waiting on the bar, you can't make any other move that turn.

You enter with a single die. Say you roll 5-2; your opponent has made the 20- and 23-points, and the 18-point is empty. You can't add the two dice together, count 7 points and come straight down on the 18-point. The checker must first enter on an open point in your opponent's home board with a single die; here both are closed, so there is no entry.

Dancing

If your opponent has made both points shown by your dice, your checker on the bar stays out that turn. When this unlucky roll happens, you are said to **dance**. Since no other checker can move while you have one on the bar, your turn is lost; you try your luck again on your next turn.

Being closed out

The worst case is when all six points of your opponent's home board are made. Then there is no entry point, whatever the dice show; this is called being **closed out**. Until your opponent breaks one of those points, your turn passes without a move every time; at the table you don't even need to roll, and the turn passes straight to your opponent. It's the same in Checkers and Dice: no dice are rolled, the screen shows "Closed out" and the turn passes straight to your opponent.

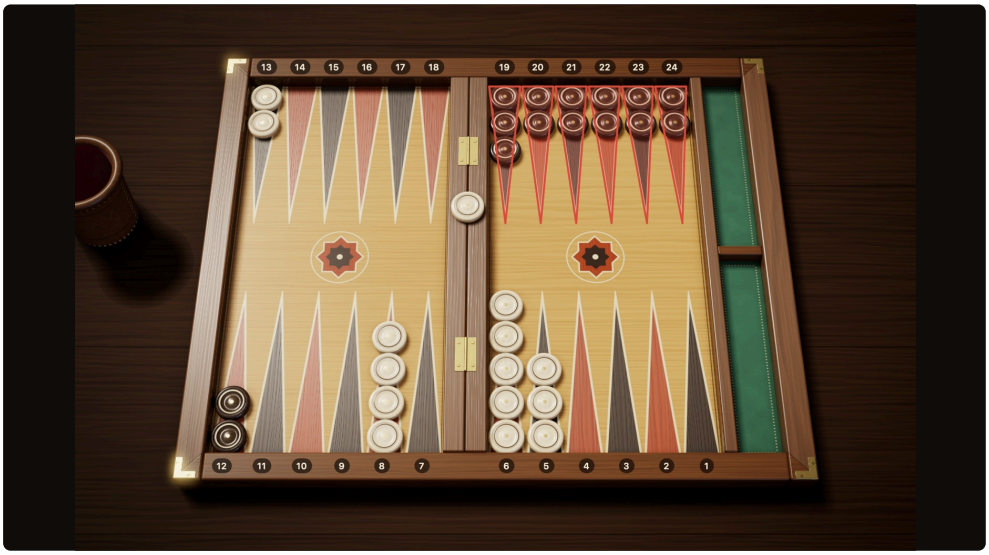


FIGURE 3.7 **Closed out: your opponent's home board is completely closed.** Your checker on the bar could enter on any open point from 19 to 24, but all six points are made (red frames). This is called being closed out: until your opponent opens one of those points, your turn passes without a roll.

3.6 Bearing off

The home stretch of the race is taking your checkers off the board; this is called **bearing off**. There is one precondition: all your checkers on the board must be in your home board, that is, on points 1–6. If even a single checker is outside it or waiting on the bar, you can't start bearing off.

The basic rule

The rule is simple: the number on the die shows which point you take a checker from. With a 6 you lift a checker off the 6-point, with a 2 off the 2-point, and put it in the space at the edge of the board. A checker that has been borne off has left the game for good.



FIGURE 3.8 **Bearing off: two checkers with 6-4.** With the 6 you take a checker from the 6-point and with the 4 one from the 4-point, and put them in the bear-off tray on the right. Bearing off can only begin when all 15 checkers are in your home board.

If there is no checker on the die's point

Sometimes the point shown by the die is empty. Then ask these two questions in order:

1. **Do I have a checker on a higher-numbered point?** If so, you can't use that die to bear off from a lower point; you play the die by moving a checker, usually one on that higher point.
2. **No checkers on any higher point?** Then you bear off a checker from the highest point on which you have one. For example, if your farthest checker is on the 4-point, either a 6 or a 5 bears that checker off.



FIGURE 3.9 **Bearing off: with a checker higher up, you move it.** For the 5: the 5-point is empty, but there is a checker on the higher 6-point; you can't use the 5 to bear off from the 3-point or the 1-point. You play the 5 by moving a checker from the 6-point (6→1), and with the 1 you bear off a checker from the 1-point. Another way: first 6→5 with the 1, then bear that checker off from the 5-point with the 5. The bear-off tray already holds 10 checkers borne off earlier.

You don't have to bear off

If you have another legal move, you don't have to take a checker off. Moving checkers forward inside your home board and leaving the bear-off for later is called **"sallamak"** in Turkish. Sometimes this is the wisest choice, so as not to leave your opponent a blot to hit. You may also move a checker with one die and bear the same checker off with the other. This freedom belongs to the international rule; in the common form of Turkish tavla, moving a checker forward with the lower die and wasting pips when you could bear a checker off is considered forbidden. In Checkers and Dice, when you play by Turkish rules, you can turn this restriction on with the "Must bear off" option in the New Match panel; it is off by default.

If you are hit while bearing off

If your opponent still has a checker waiting in your home board, a blot you leave while bearing off can be hit. Bearing off then stops: the hit checker must first enter in your opponent's home board and travel around the board back to your home board. Once all your remaining checkers are gathered in your home board again, you carry on from where you left off.



FIGURE 3.10 **Bearing off: from the highest point.** You have no checkers left on the 6- and 5-points or higher. So with each die you bear off one checker from the highest occupied point, the 4-point.

3.7 End of the game and scoring

The first player to bear off all their checkers wins the game. The size of the win, though, depends on **how far behind your opponent is** at the moment the game ends. When the game is over, look at your opponent's checkers; one of the three cases below will apply:

TABLE 3.3 Game results and points

Result	Opponent's position	International, no cube	International, with cube	Turkish tavla
Single game	Has borne off at least one checker	1	Cube value × 1	1
Gammon	Has borne off no checkers	2	Cube value × 2	2
Backgammon	Has borne off no checkers and still has a checker on the bar or in your home board	3	Cube value × 3	2 (counts as a gammon)

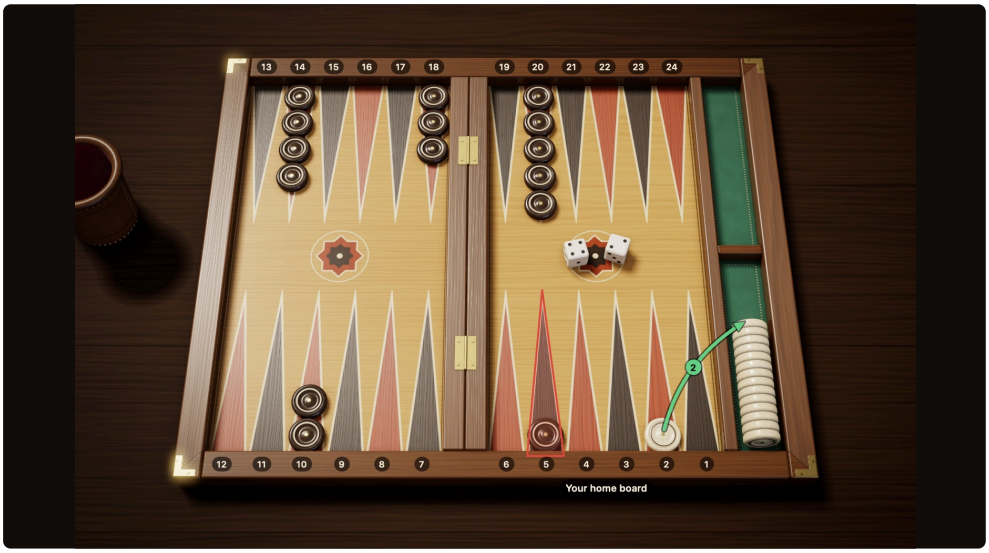


FIGURE 3.11 **Backgammon: your opponent has borne off nothing and has a checker in your home board.** You bore off your last checker with a 2 and won the game. Your opponent hasn't borne off a single checker: that alone means a gammon (2 points). What's more, they still have a checker on the 5-point, framed in red, which is in your home board: a backgammon (3 points).

Examples:

- If your opponent has borne off 4 checkers when you bear off your last one, you win a single game: 1 point.
- If your opponent hasn't borne off a single checker but has none left on the bar or in your home board, you win a gammon: 2 points.
- If your opponent hasn't borne off any checkers and still has one on the bar or in your home board, you win a backgammon: 3 points.

- With the doubling cube on 8, a player who wins a gammon scores 8 × 2 = 16 points.

In Turkish tavla a backgammon is not a separate result; that situation also counts as a gammon and is worth 2 points. Make it clear which rule set you're playing before the game.

Match score. Backgammon is usually played not as a single game but up to a set number of points; in tournaments this is usually 5, 7 or 11 points. At the end of each game, the winner's points are added to their score; when playing with the cube, the points are multiplied by the cube value. The first to reach the target score wins the match. That is why winning a gammon, or saving one, can change the outcome of the match.

3.8 The doubling cube

The doubling cube is a cube with numbers on it that is never rolled. Its faces show 2, 4, 8, 16, 32 and 64, and it simply shows the current value of the game. You rarely see it in neighborhood games in Turkey, but it is indispensable in international tournaments and online matches. Checkers and Dice has the doubling cube too; it is used when you choose the international rule set at the start of the game. Doubling and responding to a double are covered in section 5.4. If you're going to play without the cube, you can skip this section.

Where the cube sits and what it means

At the start of the game the cube sits between the two players with **64** on its top face. Since the cube has no 1, the 64 here means "not doubled yet; the game is worth 1 point." While the cube is in the middle, either of you can make the first offer. Once an offer has been accepted, the player the cube sits in front of is the one who has the right to make the next offer. In Checkers and Dice the cube is shown below the match info at the top right: at first it reads "1 Cube centred"; once a double has been taken, it shows the cube's value with "Cube yours" or "Cube theirs."

Doubling, step by step

1. **When?** You can only make an offer on your own turn, before you touch the dice. Once the dice are rolled, the chance for that turn has passed.
2. **How?** If you think your position is good, you turn the cube to the next value (2 the first time), hold it out to your opponent and say "Double." It means:

"Shall we play this game for twice the stakes?"

3. **What can your opponent do?**
   - **Take:** They take the cube to their side with the doubled value (2 on the first double) facing up; the game continues from the same position, now at twice the value.
   - **Drop:** They concede the game on the spot. The points they lose are the cube's value before the offer; on the first offer that is 1 point.
4. **Whose is the next offer?** The cube passes to the player who accepted the offer, and only that player has the right to the next double. That is why offers always alternate; the same player can't double twice in a row.

Example

Say your opponent doubled early in the game and you took; the cube is on 2, in front of you. A few turns later the game swings your way, so you turn the cube to 4 and offer it to your opponent. If they drop, you win the game and score 2 points. If they take, the cube moves in front of them on 4, and the points won at the end of the game are read from the "4" row of Table 3.4.

TABLE 3.4 Points won by cube value

Cube	Single game	Gammon	Backgammon
Centered (1)	1	2	3
2	2	4	6
4	4	8	12
8	8	16	24

What you need to know:

- There is no ceiling on doubling; after 64 the value keeps doubling to 128, 256 and so on. Still, values this high are very rare.
- If you hold the right to double, having a checker waiting on the bar doesn't stop you from making an offer. In Checkers and Dice the only exception is being closed out: while your opponent's home board is completely closed, your turn passes without a roll, and you can't double that turn either.
- Whether to double, and whether to accept, are among the most important skills in the game; guidelines follow under the next heading.

Cube decisions

Using the cube well is one of the subtlest skills in backgammon. What follows are not hard rules but sound guidelines to start with.

When to take: When you are doubled, ask yourself roughly this: "If I played this game four times, would I win at least once?" If your chance of winning is about one in four or better, taking is usually right. When you drop, your loss is certain; when you take, you keep the chance to turn the game around and later use the cube in your own favor. This 25% guideline comes from the simplest model, which ignores gammons and redoubles: owning the cube pulls the threshold down a little, the risk of a gammon pushes it up, and the match score can move it either way.

When to double: If your position is clearly better but the game isn't over yet, it's time to double. Avoid three mistakes:

- **Doubling too early:** A double offered with only a small edge gives your opponent both an easy take and ownership of the cube.
- **Doubling too late:** Your position gets so good that your opponent no longer takes and gives up the game for 1 point, whereas if you had doubled a turn earlier, they would have taken and you would be playing for twice the stakes. This is called "losing your market."
- **Too good to double:** A position can be so strong that your opponent would certainly refuse a double. If what you expect to win by playing on without the cube, thanks to your gammon chances, is more than the 1 point your opponent's drop would give you, the position is "too good to double." This guideline doesn't hold in sessions played with the Jacoby rule; there, since a gammon counts as a single game while the cube is in the middle, you need to turn the cube. In match play, too, the score changes the value of a gammon.

The 8-9-12 rule in a pure race. Once contact is over and the game is just a pip race, the pip count makes the decision. Work out the difference as a percentage by dividing it by the pip count of the player who is ahead; if you're the one behind, divide by your opponent's count, the leader's:

TABLE 3.5 The 8-9-12 rule in a pure race

Pip difference (relative to the leader's count)	Decision
8% ahead	Consider the first double
9% ahead	If you own the cube, redouble
Up to 12% behind	Take if you are doubled
More than 12% behind	Drop

Example: if your pip count is 80 and your opponent's is 88, you are 8 pips ahead; that is 10% of 80, the leader's count, and you are in doubling territory. Because the rule is based on the raw pip count, be a little more careful if you have piled your checkers on low points. The rule applies when it is your turn, once contact is completely over, and in medium or long races; when the game is down to its last few checkers, you need to look at where the checkers are rather than at percentages and, in a match, weigh the score separately.

Match and session rules

Crawford rule: The first time one player in a match comes within one point of the target (4 points in a 5-point match), the cube is not used in the very next game; this is called the Crawford game. Once that game is over, the cube comes back into use and is available until the end of the match. The rule applies only once per match; if the other player later reaches the same score, there is no new Crawford game. The aim is to stop the trailing player from doubling every game straight away and closing the gap in a single game. The rule is named after the American player John R. Crawford.

Optional rules (found in sessions, not used in matches):

- **Automatic doubles:** If the opening dice are the same, the cube is turned to 2 automatically and stays in the middle. This is usually limited to once per game.
- **Beaver:** A player who has been doubled can, while accepting, immediately double the cube once more, and the cube still stays with them. The player who doubled can take the beaver or drop; dropping costs the value of the double they offered, which is 2 points on the first double.
- **Raccoon:** The player facing a beaver, that is, the one who made the first double, can immediately double the value once more. The cube still stays with

the player who accepted the double; the other side either accepts the new value or gives up the game at the current value. Like the beaver, it is found only in sessions.

- **Jacoby rule:** While the cube is in the middle, that is, while no double has been taken in the game, gammons and backgammons count as a single game; the rule still applies after an automatic double, which leaves the cube in the middle. This encourages the leading player to double rather than chase a gammon.

CHAPTER SUMMARY

- Each die moves one checker that many points; checkers move only toward home, and the two dice are two separate moves.
- A checker can land on an empty point, on your own checkers or on an opposing blot, but not on a point your opponent has made.
- When you roll doubles, you play that number four times.
- You must play any die that can be played; if each die can be played on its own but not both together, you play the higher one.
- A hit checker goes to the bar and must first enter in your opponent's home board. A turn in which it can't enter is called dancing; when all six points of your opponent's home board are made, you are closed out.
- Bearing off begins once all your checkers are in your home board; the number on the die shows which point to take a checker from.
- If your opponent hasn't borne off any checkers, it's a gammon, worth 2 points. Under international rules a backgammon is worth 3 points; in Turkish tavla it counts as a gammon too.
- The doubling cube doubles the value of the game; the side being doubled either takes or drops.



CHAPTER 4

Playing better

The basics of playing well, once you know the rules: a sample game, pip counting, opening moves, game plans, the risk of leaving a blot and bearing-off strategy.

4.1 Sample game: the first five turns

To see the rules working together, let's follow the first five turns of a game step by step. The board is in the starting position, and we're playing by international rules. Point numbers are given from your side.

1. **Opening roll.** You roll 3 and your opponent rolls 1. You rolled higher, so you play this 3-1 as your first move. With the 3 you play 8→5, and with the 1, 6→5, making the 5-point. This is one of the most commonly recommended opening moves.
2. **Your opponent rolls 6-4.** They run one of their back checkers: the checker on your 1-point goes to 7 with the 6 and on to 11 with the 4. The checker is left alone on 11, so it is a blot.
3. **You roll 4-2.** With the 2 you play 13→11 and hit your opponent's blot; their checker goes to the bar. With the 4 you play 24→20.
4. **Your opponent rolls 5-5.** With a checker on the bar, they must enter it first. With a 5 they would have to enter on your 5-point, but you have made that

point. Since all four dice are 5s, your opponent has **danced**; they can't move a single checker this turn.

5. **You roll 6-5.** You take the checker left on the 24-point to 18 with the 6 and on to 13 with the 5. The 18-point is empty, so the intermediate point is open; you already have checkers on 13.

After five turns, the board looks like Figure 4.1:

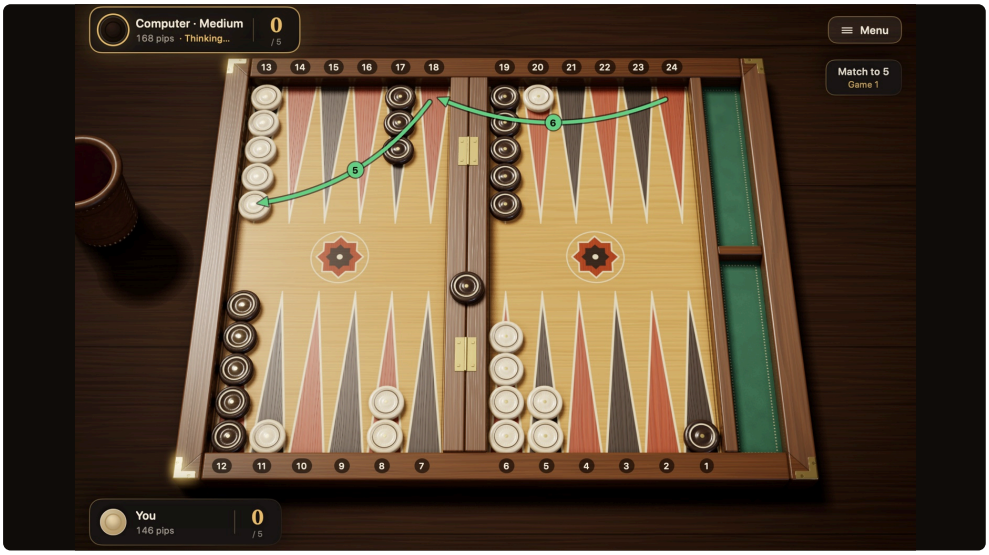


FIGURE 4.1 **Sample game: the position after the fifth turn.** Last turn: with 6-5, the checker on the 24-point ran first to 18, then to 13 (green arrows). One of your opponent's checkers is waiting on the bar; your made 5-point makes it harder for that checker to enter. Your blots are on 20 and 11. Pip counts on the player cards: you 146, your opponent 168; it's your opponent's turn.

These five turns used the rules for moving, making points, hitting, entering and dancing. The game goes on this way until the two sides have brought their checkers into their home boards and borne them off.

4.2 Pip counting

Because backgammon is a race, the question "who is closer to home?" matters a great deal at any moment in the game. The measure of this is the **pip** count: the total number of points a player's checkers must travel to reach the home board and be borne off.

How is it calculated? Each checker's distance is the number of its point in your own numbering. A checker on the 6-point is 6 pips away, one on the 13-point 13

pips. A checker on the bar is as far away as it can be and counts as 25. You add up the distances of all your checkers.

For example, at the start: 2 × 24 + 5 × 13 + 3 × 8 + 5 × 6 = 48 + 65 + 24 + 30 = **167 pips**. Both players start the game with 167. With every move your count goes down by the distance the checker actually travels (when you bear off from the highest point with a larger die, the count drops by the number of the checker's point, not by the value of the die); a checker that is hit, on the other hand, goes back to the bar and sends its owner's count up in one go.

In Checkers and Dice you don't need to work this out. The pip count on the player cards updates automatically after every move. In the sample game in section 4.1, you are on 146 pips and your opponent on 168; in other words, you are about three rolls ahead.

What is this good for?

- If you're ahead, avoiding contact and heading home quickly is usually a good plan.
- If you're behind, it's hard to win by racing alone; you watch for chances to hit your opponent's checkers or block their path.
- If you're playing with the cube, the pip difference determines your cube decisions in a pure race. What to do at which difference is explained under the heading "Cube decisions" in section 3.8.

4.3 Opening moves

Since the first move of the game is always made from the same position, the good plays for each roll have largely been settled by computer analysis. Table 4.1 gives the widely accepted plays for games under international rules in which the match score is not taken into account; in a match, the score can change some choices. The notation is "from/to": 8/5 means from the 8-point to the 5-point; (2) shows that the same move is made with two checkers.

TABLE 4.1 Opening moves under international rules

Roll	Recommended play	Reasonable alternative	The idea
 3-1	8/5 6/5	—	Making the 5-point
 4-2	8/4 6/4	—	Making the 4-point
 6-1	13/7 8/7	—	Making the 7-point, blocking the way in front of your home board
 5-3	8/3 6/3	—	Making the 3-point
 6-5	24/13	—	Bringing one back checker to safety in a single move
 6-4	24/14	24/18 13/9 or 8/2 6/2	The three plays are almost equal in value
 6-3	24/18 13/10	24/15	Developing from both the back and the middle
 6-2	24/18 13/11	24/16	Developing from both the back and the middle
 5-4	24/20 13/8	13/9 13/8	A potential anchor at the back, a new checker in the middle
 5-2	13/8 24/22	13/8 13/11	Safe development from the middle
 5-1	13/8 24/23	13/8 6/5	A safe checker, splitting the back checkers
 4-3	13/10 13/9	24/21 13/9 or 24/20 13/10	Bringing down builders for your home board
 4-1	24/23 13/9	13/9 6/5	Splitting the back checkers, a new checker in the middle
 3-2	24/21 13/11	13/11 13/10	Splitting the back checkers, a new checker in the middle

Roll	Recommended play	Reasonable alternative	The idea
2-1	13/11 6/5	13/11 24/23	Slotting the 5-point or splitting the back checkers

The plays in Table 4.1 are based on the computer analyses on the Backgammon Galore website. The plays in the "Reasonable alternative" column are good too; in the analyses the difference between them is very small, and you can choose according to your style of play. Beginners who want to play safe may prefer 13/11 24/23 for 2-1 and, for 6-4, 8/2 6/2, which leaves no blot at all. For comparison: the blot left by 24/14 is hit by 11 of the 36 rolls; at least one of the two blots left by 24/18 13/9 is hit by 27 of the 36.

Under the international rule, the first move is never doubles. If you're playing by the Turkish rule, the first roll can also be doubles; in that case these plays make a good start:

TABLE 4.2 Opening doubles under the Turkish rule

Doubles	Recommended play
6-6	24/18(2) 13/7(2)
5-5	13/3(2)
4-4	24/20(2) 13/9(2)
3-3	8/5(2) 6/3(2)
2-2	13/11(2) 6/4(2)
1-1	8/7(2) 6/5(2)

You don't need to memorize these tables. But it's instructive to see the logic they share: if possible, make a new point in your home board or just in front of it; if you can't, get your back checkers moving, and don't leave single checkers where they are easy to hit.

4.4 Basic game plans

Most backgammon games turn into one of a few familiar plans. Which one suits you depends on the pip difference and the points made on the board. It's also

normal to switch from one plan to another as the game goes on.

- **Running game:** If you're ahead in the pip count, you break contact with your opponent and bring all your checkers home quickly. Once contact is over, the game becomes a pure dice race, and the player ahead usually wins.
- **Holding game (anchor):** You hold a point in your opponent's home board with two checkers. This "anchor" protects you from being hit and gives you a chance to hit your opponent if they leave a blot on their way home. If you're behind, it's the soundest plan.
- **Priming game:** By making points side by side, you build a barrier in front of your opponent's back checkers. If six points are made in a row, a checker behind them can't get past with any roll; you slowly move the prime toward your home board and squeeze your opponent.
- **Blitz:** You hit your opponent's checkers again and again as they enter your home board, and close your home board fast. A successful blitz closes your opponent out and often wins a gammon; but if it fails, you leave blots of your own behind.
- **Back game:** When you're far behind, you hold two anchors in your opponent's home board and wait for them to leave a blot as they come home. It's a rescue plan, not a starting point you would choose on purpose.

These are rules of thumb, not mechanical rules. Timing, the quality of your points, the state of your back checkers and the risk of a gammon can call for the opposite of what the pip difference suggests: giving up a strong prime right away just because you have a big pip lead is a mistake, and so is ending a blitz that's going well just because you're behind.

4.5 The risk of leaving a blot

Sometimes you can't avoid leaving a checker on its own. Then the question is: where can I leave it so that it's least likely to be hit? The answer depends on how many points away the blot is from the opposing checker that could hit it.

There are three practical lessons to draw from Figure 4.2:

1. **The most dangerous distance is 6.** A blot 6 points away is hit by 17 of the 36 rolls, almost one roll in two.
2. **7 and beyond is much safer.** To hit a checker at this distance, both dice have to work together; the risk drops from nearly one in two to one in six and

below.

3. **Close is not always safe.** Even a blot 1 point away is hit by 11 of the 36 rolls.

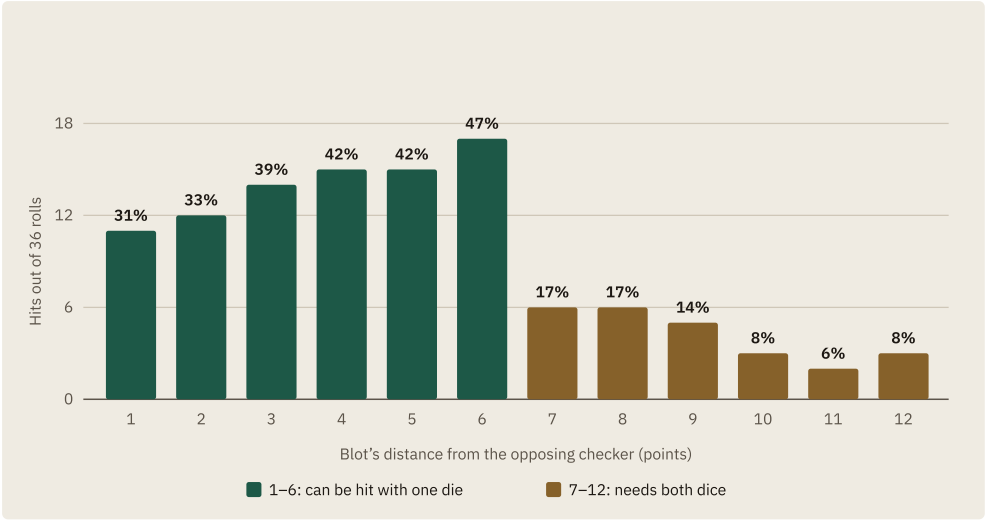


FIGURE 4.2 **The chance of a blot being hit, by distance.** A blot 6 points away is in the most danger; from 7 on, the risk drops sharply. The bars show how many of your opponent's 36 possible rolls hit your blot (assuming no made points in between).

These numbers assume a single opposing checker, a path with no made points in between and no opposing checker waiting on the bar. If more than one opposing checker bears on the same blot, the risk goes up; if your made points block the intermediate points, it goes down. Because of doubles, distances such as 12, 16, 20 and 24 can also be reached in special ways; the risk does not fall smoothly with distance.

4.6 Bearing-off strategy

The bearing-off rules are simple, but bearing off well is a skill. From the same position, two players making different decisions can end up a roll or two apart; in a close race, that difference decides the game.

Bringing checkers home

- **Bring checkers in on the high points of your home board.** As checkers come in, try to place them on the 6-, 5- and 4-points. Checkers piled up early on the 1- and 2-points will make you waste big numbers later.

- **Spread them evenly.** Spreading your checkers across the points lets you bear off a checker whatever you roll. Empty points, on the other hand, mean you can't bear off a checker when that number comes up.
- **Don't go deep for no reason.** Taking a checker that has come home all the way to the 1-point, just to "move it on a bit," means that once bearing off starts, a big number like a 6 is worth only 1 pip.

With no contact

Once your opponent's checkers have passed all of yours, there is no longer any chance of being hit and the game is a pure race. The basic principle then is: **bear off as many checkers as you can every turn.** Computer analysis shows this is the best play in almost every position; only with very uneven distributions can it be better to rearrange a checker inside your home board so that later rolls are wasted less. If you can't bear a checker off, use the die to fill empty points. If you roll doubles, don't miss the chance to bear off four checkers at once.

With contact

If your opponent still has one or two checkers in your home board, bearing off is not a speed race but a safety game. A single blot gives your opponent a chance to turn the game around.

- **Clear from the back.** Clear the checkers on your highest points first. Checkers waiting close to an opposing checker are the most dangerous ones.
- **Keep even numbers of checkers.** If you keep checkers in twos and fours on your points, you can clear them two at a time with both dice and empty a point without leaving a single checker behind. On a point with an odd number, one checker is often left on its own.
- **Try to fill the gaps.** Empty points in your home board can force you to leave a blot on later rolls.
- **Move within your home board if you have to.** If bearing a checker off would leave a blot, it's often better to shift checkers inside your home board and stay safe.

The same principles apply if you have started bearing off while an opposing checker you hit is waiting on the bar: your opponent can come in and hit your blot at any moment, so keep your home board as closed as you can while you bear off.

4.7 Tips for beginners

Once you've learned the rules, these simple principles will help you in your first games:

1. **If you must leave a blot, choose where.** A single checker is sometimes unavoidable; then leave it where your opponent can hit it with the fewest rolls, preferably 7 or more points away.
2. **Value the points in your home board.** Every point you make in your home board makes it harder for an opposing checker you've hit to come back in. The 5- and 4-points are the most valuable.
3. **Don't stack your checkers in one place.** Piling five or six checkers on one point means those checkers sit there doing nothing. Take checkers from the stacks and use them to make new points.
4. **Stay flexible.** Checkers standing at different distances make most future rolls useful. Avoid a setup that depends on the same numbers.
5. **Think from your opponent's side too.** Before you move, take a quick look at what your opponent could do with their next roll.
6. **Don't rush.** Compare a few options for every roll. In Checkers and Dice a step you've played can't be taken back by default. If you want to try different moves, turn on "Show the Undo button" and turn off "Auto-confirm moves" in Settings; details in section 5.3.
7. **Learn from the games you lose.** In backgammon, luck can decide a single game; but if you keep making the same mistake, the problem isn't the dice.

CHAPTER SUMMARY

- The pip count is the total number of points your checkers must travel before they can all be borne off; both sides start the game with 167 pips.
- Opening moves have largely been settled by computer analysis. The common logic is to make a point in or in front of your home board or, if you can't, to get your back checkers moving.
- Games usually turn into one of a few plans: running game, holding game, priming game, blitz and back game.
- If you have to leave a blot, the most dangerous distance is 6; 7 points or more is much safer.
- With no contact, bear off as many checkers as you can every turn; with contact, bearing off is not a speed race but a safety game.



CHAPTER 5

Using Checkers and Dice

This chapter answers, step by step, the question of how to do things in Checkers and Dice, from setting up a match to the settings and the practice tools. Button and menu names are given exactly as they appear in the app.

5.1 Starting a new match

On the main menu, press “New Match” (⌘N). The panel that opens has five choices; “Start” starts the match.

1. **Difficulty:** “Easy” makes frequent mistakes and is ideal for learning. “Medium” is a trained neural network that makes the odd mistake. “Hard” is a neural network trained on millions of games. “Master” also calculates every possible roll. “Adaptive” picks its strength from the game grade of your recent games or, if you have no analyzed games, from your rating (section 5.7); the line below the chips shows which level that corresponds to, and the strength stays the same for the whole match.
2. **Opponent style:** “Balanced,” “Aggressive,” “Cautious” or “Runner.” A balanced opponent plays the best move. The other three choose by their own leaning among moves that differ only very slightly: the aggressive one prefers hitting and making points, the cautious one prefers not leaving a checker exposed, and the runner prefers escaping its back checkers; the cost

of this is small. The style affects only the computer's choice of move; the hint, the Coach and the analysis always rely on the best move. The chosen style is saved with the match.

3. **Rules:** "Turkish tavla" or "International." The differences are in Table 2.1. With Turkish tavla selected, two traditional rules appear below: "No hit-and-run" and "Must bear off." Both are off by default. The "Clock" chip at the end of the row turns on the match clock: in each game, each side gets twice as many minutes as the match length (2 in a 1-point match, 30 minutes in a 15-point match; 15 minutes in a session), plus a delay of 12 seconds per move; the time doesn't start to run down until the delay has passed. A side whose time runs out loses that game at the cube value. The time left is shown on the player cards and turns red in the last 30 seconds. The rules and the clock you choose stay the same for the whole match.
4. **Match or session:** With "Match (points)" you choose how many points the match is played to: 1, 3, 5, 7, 9, 11, 13 or 15. Under international rules there is also a "Session" tab next to it: a session has no match length, each game's points are recorded, and you end the session whenever you like. In a session the "Jacoby rule," "Beaver," "Raccoon" and "Automatic doubles" options also appear; what they are is explained under the heading "Match and session rules" in section 3.8. Turkish tavla has no sessions.
5. **Play as:** "Light" or "Dark." Whichever you choose, your checkers are always at the bottom. The icon's color comes from the checker style you chose in Settings: with "Red & Black Lacquer," the light checker is red.

If you start a new match while one is in progress, the app asks first. The match in progress is saved; you can go back to it from the "Matches" card. The confirmation card also shows a summary of the match about to start. ⌘N brings up the same confirmation, between games too.



FIGURE 5.1 **New match panel: difficulty (including "Adaptive"), opponent style, rules and clock, match or session and checker color.**

5.2 Rolling the dice and moving checkers

When it's your turn, the dice are rolled for you: they pour out of the leather cup onto the right half of the board. Under international rules, if you have the right to double, the app waits; the "Roll Dice" button appears in the right half of the board, with the "Double" button below it. You roll by pressing the button, clicking the dice or pressing the Space bar. If "Roll automatically" is off in Settings, the "Roll Dice" button appears on every one of your turns. If "Play with real dice" is on, you roll real dice on the table yourself: a panel opens in place of "Roll Dice," you enter the faces of the two dice with its buttons or the 1–6 keys and press "Done" (Return), and the dice pour out of the cup showing the faces you entered. You also roll and enter the computer's dice; for the opening roll you are asked for your die first, then the computer's. In this mode there are no verifiable dice (section 5.8).

After the dice are rolled, the checkers you can move glow. You move a checker in one of these ways:

- **By clicking:** First click the checker; the points it can move to are marked with green rings. Then click the target. A target where you would hit your opponent's blot is marked with a red crosshair.
- **By dragging:** Pick up the checker and drop it on the target.
- **Quickly:** A double-click plays the checker with the higher die, a right-click with the lower die.
- **With the keyboard:** Tab moves between the checkers you can move, the arrow keys choose the target and Return plays the move.



FIGURE 5.2 **Your turn to roll, when you have the right to double.** 1 the "Roll Dice" button, 2 the "Double" button, 3 the cube indicator.

To change which die you play first, click the dice on the board. Once you play your last step, your turn ends by itself and passes to your opponent; if you have only one legal move, the app plays it for you. If you'd rather confirm your turn yourself or take steps back, turn off "Auto-confirm moves" and turn on "Show the Undo button" in Settings: then "Done" (Return) ends your turn, and until then "Undo" (⌘Z) takes back the steps you've played, one at a time. If you have no move at all, the app says "No legal moves!" and the turn passes to your opponent. If you have a checker on the bar and your opponent's home board is completely closed, the dice aren't even rolled: the screen shows "Closed out" and the turn passes straight on.



FIGURE 5.3 **Making a move.** The roll is 4-2 and the checker on the 13-point is selected. 1 the selected checker, 2 the empty point it can reach with the 4 (green ring), 3 the point where it can hit your opponent's blot with the 2 (red crosshair).

5.3 Hints and gameplay aids

The "Hint" button is hidden by default; it appears at the bottom right once you turn on "Show the Hint button" in Settings. The button (⌘I) marks on the board the move the Master engine suggests for that turn; the marking disappears once you play your first step. Under international rules it also gives advice on cube decisions: before you roll, "Suggested: double" or "Suggested: no double, just roll"; when the computer doubles, "Suggested: take" or "Suggested: drop." The hints you use are recorded in the end-of-game summary and in the statistics.

"Coach" is a learning mode; it is off by default in Settings. When it is on, the Master engine compares the move you played with the best one every time you finish a turn. If your move loses clearly against the best option, the turn does not end and a card appears: the size of the mistake in points, the move you played and the better one, and the better move's arrows on a board diagram. "Undo" takes back all the steps of the turn (even when the "Undo" button is hidden) and you play again with the same dice; "Play Anyway" ends the turn. Under international rules it also checks cube decisions: if you want to roll when you should double, it warns you and you can double right from the card; if you

double when you shouldn't, or answer the computer's double wrongly, it warns you and you either keep your decision or think again. On the card, Esc chooses the first option ("Undo," "Cancel" or "Think Again"); on the card that says you should double, however, it only closes the card: no double is offered and no dice are rolled, and the decision is still yours. "Warning threshold" chooses when the warning comes: "All mistakes" warns for losses of 0.02 points or more, "Only blunders" for 0.08 points or more. If the Coach warned you, the count appears in the end-of-game summary. The end-of-game analysis still runs; because it also tests the move by simulation, the loss there may differ slightly.



FIGURE 5.4 **Coach card.** The move played is 0.09 points behind the best; the arrows in the diagram show the better move, and the line with the bulb icon gives the reason.

The Coach card, the move analysis, the review panel and the share card also say in one sentence why the better move is better; the sentence sits on the line that starts with a bulb icon, for example "The best move hits a checker and makes the 5-point." The app compares where the two moves leave the board and counts only the differences in the best move's favor, at most two of them: hitting, the number of blots and shots left, a new made point or anchor, the length of the prime, back checkers escaping or the opponent's being held, made points in the home board, the number of checkers borne off, gammon chances and gammon risk. If another candidate is selected in the review, the sentence is built relative to that move. If "Natural-language explanations" is on in Settings, the on-device language model turns this sentence into more fluent prose on the Coach card, in the move analysis and in the review panel: the template sentence

appears first, and when the model's sentence takes its place, a sparkle icon appears where the bulb was. Numbers and terms stay exactly as they are, and nothing leaves the device. The option is available on devices with Apple Intelligence; it is on by default in English, French, German, Spanish, Portuguese and Italian, off in Turkish, and not available in Greek and Russian.

The "Gameplay" section of Settings has options that make your turn easier; all of them are on by default except the "Hint" and "Undo" buttons, "Play with real dice" and "Coach" (in Turkish, "Natural-language explanations" is off too):

- **Roll automatically:** When it's your turn, the dice are rolled for you. On turns when you could double, it waits; the cube decision is yours. When this is off, you roll every turn yourself with the "Roll Dice" button or the Space bar.
- **Auto-confirm moves:** Your turn ends by itself with your last step. When this is off, you end your turn with the "Done" button (Return); if "Undo" is on, you can take steps back until then.
- **Play forced moves automatically:** If there is only one legal move, the app plays it and your turn ends.
- **Play with real dice:** Off by default. You roll real dice on the table yourself and enter the faces; you enter the computer's dice too. Details in section 5.2.
- **Highlight movable checkers** and **Show target rings:** Turn the glowing checkers and the green rings on or off. With the rings off, a target ring appears only on a target you pick with the keyboard.
- **Show position type:** Shows the type of position in the match info: opening, running game, bear-off, blitz, back game, priming game, holding game and others; game plans are in section 4.4.
- **Show the Hint button** and **Show the Undo button:** Both are off by default: you can't ask for a suggestion or take back a step you've played. When you turn them on, the buttons appear at the bottom right and the ⌘I and ⌘Z shortcuts work too. While "Undo" is off, a suggestion can be asked for only before your first step.
- **Coach:** Off by default. When you turn it on, a "Warning threshold" option appears below it: "All mistakes" or "Only blunders." Details above.
- **Natural-language explanations:** Turns the move explanation into a fluent sentence with the on-device language model; details above.

5.4 Doubling and responding to a double

You can double only under international rules. When it's your turn and the cube is in the middle or yours, press the "Double" button (⌘D) before you roll. The button shows the value the cube will go to, for example "Double ×2." The computer answers at once: if it takes, the cube passes to it and the game goes on; if it drops, you win the game.

When the computer doubles, a card appears on the screen:

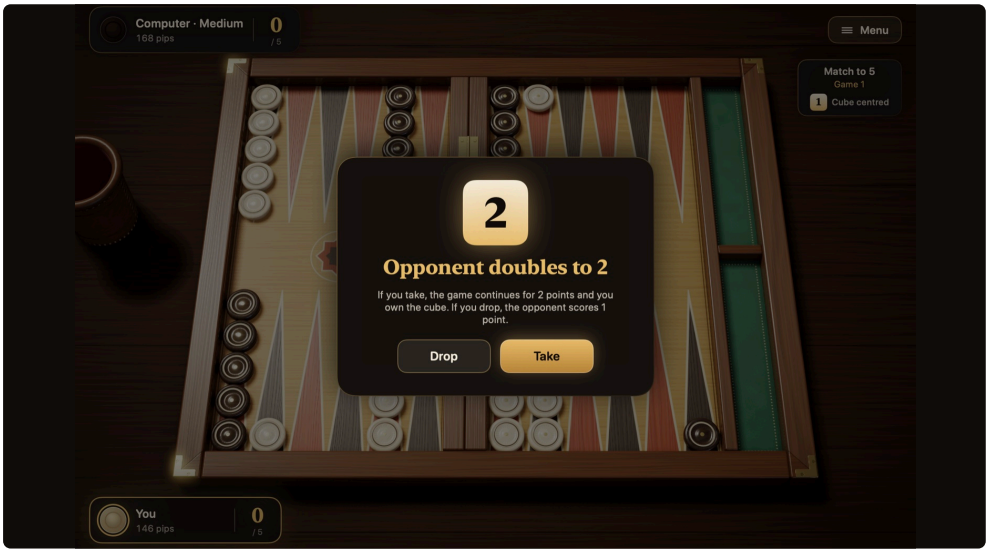


FIGURE 5.5 **The cube offer card.**

- **Take:** The game goes on at twice the value, and the cube passes to you.
- **Drop:** You give up the game; your opponent scores as many points as the cube was worth before the offer.
- **Beaver:** Appears in a session when "Beaver" is on. You take the double and immediately turn the cube one level higher, and the cube stays with you.
- **Hint:** Appears if "Show the Hint button" is on in Settings; shows the Master engine's suggestion for your decision.

If you double and the computer beavers, a similar card appears: you take, you drop or, if "Raccoon" is on, you turn the cube one level higher again.

The cube's value and who owns it are shown below the match info at the top right: "Cube centred," "Cube yours" or "Cube theirs." In a match, the cube is not used in the first game played once one side is 1 point away from the target, and that spot then reads "Crawford game."

Score awareness. In a match, the value of a decision depends on the score. You open the "Match Equity" card from the pause card, on the Mac from the Game menu, or from the "Score awareness" tile on the "Practice" card. The table gives your chance of winning the match at every score: the rows show the points you still need, the columns the points your opponent still needs, and your current score is in the gold cell. While a match or the daily position is in progress, the "This score" section at the top of the card shows your chance of winning the match at the start of the game, the gammonless take point and the gammon price; in special situations such as the Crawford game, short notes appear. There is no table in a session.

5.5 Resigning and ending a session

During a game, choose "Resign" from the Game menu or the pause card. On the card, you propose how many points to resign for: single, gammon or, under international rules, backgammon. Below each option, the card tells you in advance whether your opponent will accept. Your opponent refuses an offer below what the position promises, and the game goes on; the match score and the cube are taken into account too. Levels that can't happen don't appear on the card: once you've started bearing off, a gammon or backgammon is no longer possible, and in a match you aren't offered anything above the level that already wins your opponent the match.

The computer may also resign on its own in a hopeless race.

A session can only be chosen under international rules; Turkish tavla doesn't have one. A session has no match length; you end it yourself with "End Session." If you end it between games, from the end-of-game card, the session ends at once. If you want to end it during a game, you resign the game in progress; if your opponent accepts, the game's points are recorded and the session ends. The session's result goes into the archive and the statistics.

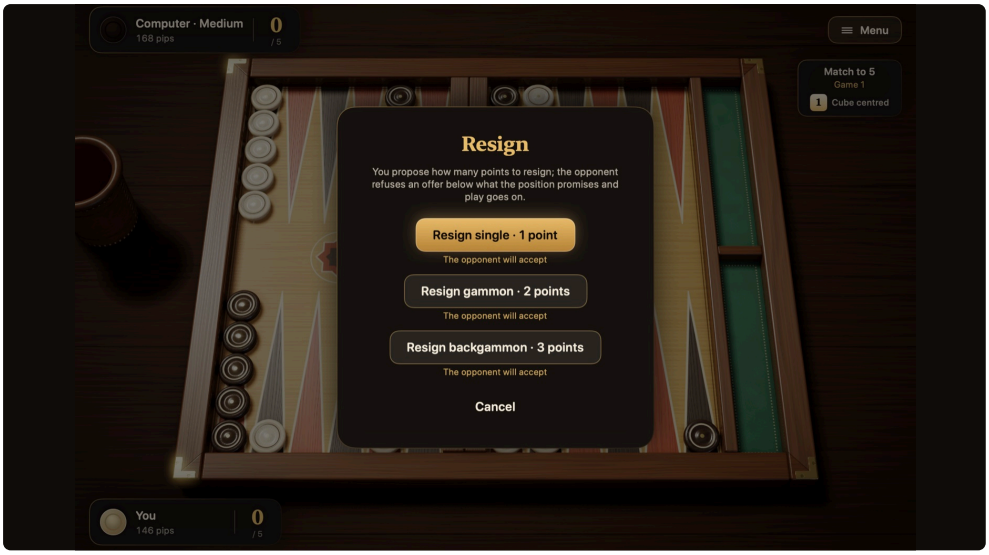


FIGURE 5.6 **The Resign card.**

5.6 Pausing and resuming a match later

The “Menu” button at the top right, or Esc, pauses the game; press Esc again and the game resumes. The pause card has “Resume,” “Settings,” “How to Play,” “Move History,” “Matches,” “Match Equity” (section 5.4), “Resign” and “Leave Match.” At the bottom, “Copy IDs” copies the XGID of the current position and its position and match IDs in GNU Backgammon format to the clipboard, and “Share” shares the position as an image; both are described in section 5.8. “Save to Library” adds the position to the library (section 5.12).

Matches are saved automatically. With “Leave Match” you can go back to the main menu and quit the app; “Continue” on the main menu reopens the last match where you left off. If you have more than one match, the “Matches” card lists the matches in progress and an archive of finished matches. You return to a match in progress with “Resume,” remove one you don’t want with “Delete” and watch a finished match again with “Replay,” starting from its first game; the move history button opens its games and turns so you can replay from the game you choose. Each row also shows the match’s rule set and any optional rules that are on. The card’s “Inspect Position from Clipboard” row sets up on the board a position ID you have copied in GNU Backgammon format (with the

match ID, if any), or an XGID, when you press “Show on Board”; the Master engine evaluates the position, see section 5.8.

With Handoff you can carry on a match in progress on your other device: a match on your Mac opens on your iPad, and one on your iPad on your Mac, right where you left off. When both devices are on the same Apple Account with Bluetooth and Wi-Fi on, and it is your turn, the Checkers and Dice Handoff icon appears in the other device’s Dock or App Switcher; tap or click it and the match is set up there from the start of that turn, with the same dice, while the copy here is parked on the main menu and stays saved. iCloud sync isn’t needed; the operating system does the transfer.

The “Continue” widget shows the score of your latest match in progress, its level and which game you are in, on the Home Screen or on the desktop; tap or click it and the match picks up where you left off, or, if no match is in progress, the New Match panel opens. Shortcuts and Siri work too: the actions “Open the Daily Position,” “Start a New Match,” “Resume the Match” and “Open Practice” appear in Shortcuts, and saying “Resume my match in Checkers and Dice” to Siri is all it takes. Spotlight finds the matches in the archive and the positions in the library; choosing one opens that match’s move history or the library. The menu bar item (Mac) and the Live Activity (iPad), which you turn on in the “System” section of Settings, also show the match in progress (section 5.9).

Matches played in other programs can enter the archive too: the “Import…” button on the “Matches” card reads Jellyfish and XG .mat files, XG’s own .xg files and GNU Backgammon .sgf files. You can also open these files directly: on the Mac with “Open With” in the Finder (or by double-clicking, if no other program is the default) or by dragging and dropping them onto the window, and on iPad from the Files app; once a file opens, the “Matches” card appears. Since the file does not say which player you are, the app shows the two players’ names and asks “Which player are you?”; results and analysis are shown from that player’s point of view. A game that contains an illegal move is cut off at that turn, and the number of games cut is reported. An imported match is listed in the archive with the “Imported” tag and the names from the file, and can be replayed with “Replay”; when the archive is full and older matches are deleted, imported ones stay. Every game in the archive, your own matches included, can be analyzed later by the Master engine with the “Analyse” button in the move history; mistakes, candidates and dice luck are reviewed as in section 5.7. “Analyse All” on the “Finished matches” heading goes through every game not yet analyzed, one after another, in the background; the progress is shown on the

same line, and the × button stops it. Once a match has been analyzed, its row shows an approximate PR value; the Statistics card gathers these values into the PR trend (section 5.7).

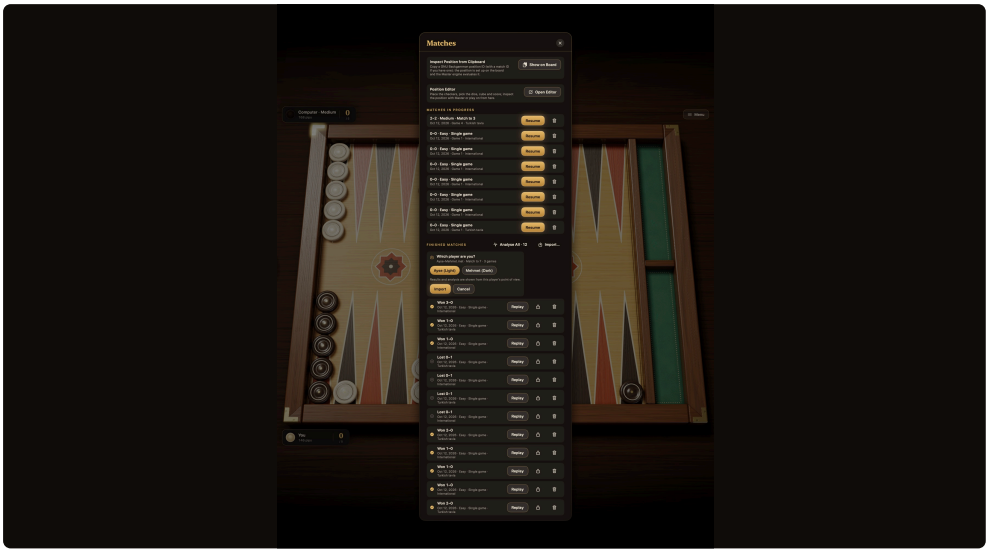


FIGURE 5.7 **Importing.** A .mat file has been read on the “Matches” card: you are asked which of the two players in the file you are; below, the matches in the archive.

5.7 End of game, analysis and statistics

When a game ends, a card shows the result: the points won, the reason for a gammon or backgammon, the match score and a summary of the game. The summary lists the time, both sides’ average roll, dice luck, the hints you used and, if the Coach warned you, how many times; dice luck is computed once the analysis is done, from the actual value of the rolls in their positions (below). When the match is over, “New Match” on the card starts a new match with your last choices, without opening the panel; a summary of the match about to start appears below the button.

The “Game grade” in the middle of the card comes from the Master engine’s review of your decisions in the game; next to it are an approximate PR value and the number of mistakes. “Details” opens the analysis card. The “Dice luck” section at the top gives both sides’ total luck, the net luck, and your luckiest and unluckiest rolls. A roll’s luck is the difference, in points, between the value of the best move that roll allows and the average over the 21 possible rolls; 0.3 points or more counts as lucky or unlucky, 0.6 or more as very lucky or very

unlucky, and is marked with an icon next to the dice in the move history. Below, the decisions that cost 0.02 points or more against the best option are listed: a small diagram shows the position at the moment of the decision and the arrows of the better move, with the move you played, the better one and the loss beside it; losses of 0.08 points or more are shown as blunders, in red and with a "!!" mark. "Candidates" opens up to the five best moves for that turn with their losses. Cube decisions are included too. When the match is over, a grade for the match is also given. "Add to Flashcards" turns the checker-play mistakes in the list into flashcards that you study with spaced repetition; blunders become flashcards automatically once the analysis is done (section 5.12).

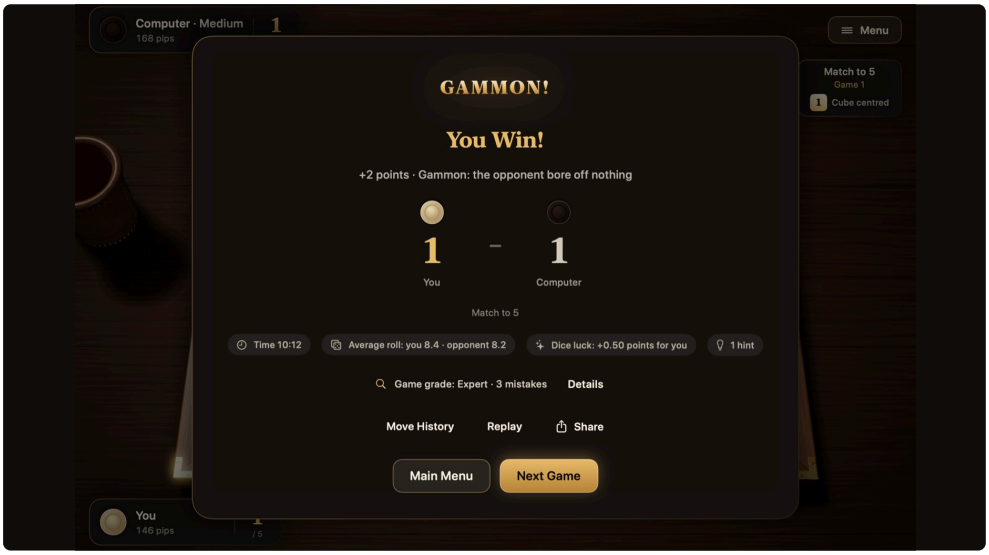


FIGURE 5.8 **The end-of-game card.**

"Show on Board" takes the decision to the board: the position is set up as it was at that moment, the best move is marked with checkers and target rings, and the review panel opens: to the right of the board in a landscape window, below it in a portrait window. Among the panel's chips, "Position" returns to the marked position; "Best," "Played" and the other candidates animate that move on the board, and each chip shows the move's loss against the best. For cube decisions the chips show the options and their losses; nothing is played on the board. With the arrows, or ← and →, you go to the previous and next mistake; with Space you cycle through the position, the best move and the move you played. "Copy IDs" and "Share" take the position outside, both in section 5.8; "Save to Library" adds the position to the library (section 5.12); "Back to Analysis" returns to the card and "Close" ends the review. "Copy Link" copies

the position as a link (section 5.8); “Similar Positions” finds the positions in your archive and in the library that are most like this one; and “Play From Here,” if you entered the review from the main menu or after the match ended, starts a free game from that position with the same dice. The panel’s subtitle also shows the decision’s position type.



FIGURE 5.9 **Move analysis.**



FIGURE 5.10 **Review panel, after Deep Review.** The move played is 0.04 points behind the best; the “Position” chip is selected and the best move is marked on the board; the line with the bulb icon gives the reason, and the line above the “Full Rollout” button shows the number of trials and the noise margin.

Similar Positions. The card lists up to twelve positions, those that are at least 60% similar: each row has a small diagram, the similarity percentage, the position type, the date, game and turn of the match the position comes from (or its name in the library) and the move played on that turn. “Show on Board” opens the position in a review. The more matches build up in the archive, the more results there are; the search also works for a position in the library (section 5.12).

The “Deep Review” button on the panel re-evaluates that decision’s candidates with far more simulation: a simulation of 6912 trials per candidate is added to Master’s 3-ply search, and the two are averaged. The simulation runs in two independent seed halves, in blocks; the noise margin computed from the spread of the blocks is shown beside the result. The losses and the order of the candidates are updated to the new values; if the difference between two candidates is smaller than the noise margin, both are playable. It takes a second or two on a Mac and a little longer on an iPad; progress is shown in the panel.

The “Full Rollout” menu on the panel plays the candidates out to the end of the game. In the menu you first choose the ply at which moves are picked, then the number of trials: “1-ply” is fast, and the moves in the simulation are picked by the fast network’s direct evaluation (648, 1296, 2592, 5184 or 10368 trials); “2-ply” looks at the opponent’s 21 possible rolls at every move and takes about three times as long (648, 1296 or 2592 trials). All the candidates are played with the same dice sequences; because variance reduction subtracts the luck of each roll from the result, a narrow noise margin is reached with few trials. The result line shows the number of trials, the ply and the largest noise margin among the candidates; the losses and the order are updated to the new values. Deep Review is faster; Full Rollout is for when you need precision. The rollout runs in a queue: it continues from where it left off if you close the review, move to another decision or quit the app; when it finishes, the result is ready the next time you open the same decision. The panel shows an estimate of the remaining time and a cancel button. On the iPad the rollout only advances while the app is in the foreground.

In a match or a session, the “Full Rollout” menu is available for cube decisions too: when you review a doubling decision or a take/drop decision, both branches of the decision (no double, and double then take) are played out to the end of the game with the same dice sequences. In the simulation both sides use the cube, and the drop is computed exactly; the ply and trial options are the same as for move decisions. When it finishes, the options’ losses are updated and each

option's value is shown beside it. On a Mac it takes a few seconds at 1-ply with 648 trials and about twenty seconds at 2-ply; on an iPad roughly twice as long. In a session, this menu isn't offered for beaver and raccoon responses.

With the buttons at the bottom of the end-of-game card, you open the move history, replay the game, share the result as an image, return to the main menu or go on to the next game.

The "Statistics" card on the main menu shows the numbers of matches and games won and played, your win rate, the gammons you've won and conceded, your best winning streak, results by level, the dice distribution (how often each face came up, with the expected count per face), your PR trend (the approximate PR values of the analyzed matches in the archive as a line from oldest to newest; below it, the latest value, the best and the average of the last five matches), the daily position results (separately for each rule set: positions solved, streak, best streak and average loss; section 5.12) and recent matches. The "Rating" section shows your local Elo-style rating, calculated from your games against the computer (it starts at 1500; the levels are rated Easy 1200, Medium 1500, Hard 1700, Master 1900), your tier ("Apprentice," "Journeyman," "Master Candidate," "Master," "Grandmaster") and a ladder of the last six months. "Your most frequent mistakes" lists up to three position types where you lose the most points (at least 20 decisions per type), with the number of mistakes and decisions and the PR; if your last ten games are better than the earlier ones, it says "Improving," and if they are worse, "Getting worse." The daily position section also holds the counters of the opening quiz and the daily cube decision; in the "By level" section, "Adaptive" has a row of its own. Sessions played and net points in sessions are kept separately. "Reset Statistics" lets you start over.

The card's "Achievements" section lists twenty badges: first win, ten and a hundred wins, first match, a match of 7 points or more, gammon, backgammon, a game won with the cube at 4 or higher, a game and a match against Master, an error-free game, a low PR, a five-game winning streak, a hundred games, solving the daily position at the first attempt and on seven and on thirty days in a row, importing a match file, analyzing a game from the archive and putting a decision through Deep Review. Earned badges appear in gold with their date, the others faded with their condition; when you earn an achievement, a short notice appears at the top of the screen. With iCloud sync on, the Statistics card and the achievements show the total across your devices: each device writes its own numbers, the card adds them all up and the daily position streak is shared;

"Reset Statistics" clears only this device's numbers. The app has no server of its own and asks for no account.

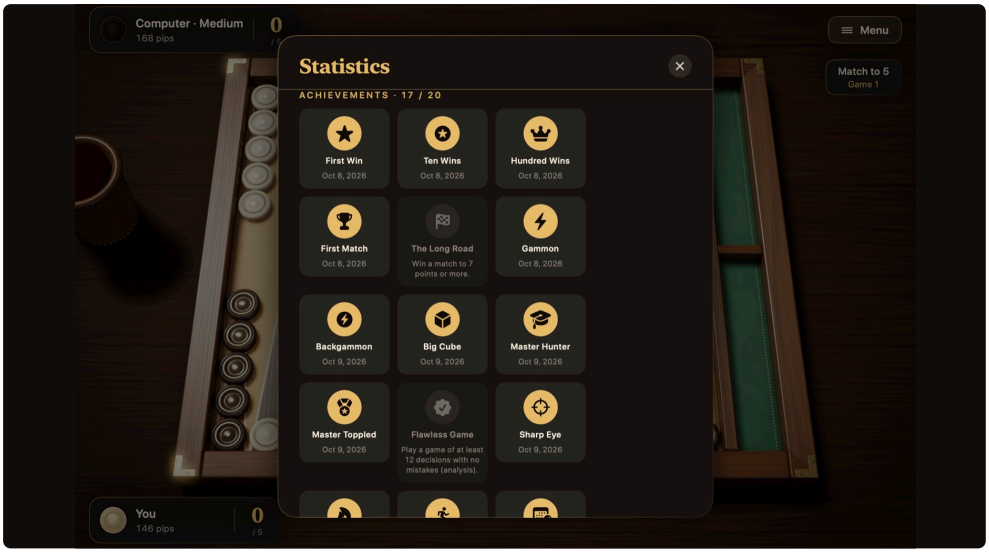


FIGURE 5.11 **The Achievements section of the Statistics card.** Earned badges in gold with their date, the others faded with their condition.

5.8 Move history, replay and export

"Move History" (⌘L) lists every turn of both sides, along with doubling and resignation events. If the match has more than one game, you choose the game with the buttons at the top. The line below the summary lets you verify the dice: each game's dice are derived from a secret seed fixed at the start of the game; while the game is in progress the line shows the seed's commitment with a lock icon, once the game is over the seed itself, and "Verify" checks that every roll in the record came from that seed. Games played with real dice, imported games and older games don't have this line.

With "Replay" you can watch a finished game from the beginning with the real animations: ← and → step between turns, and Space plays and pauses. If the match is over, or if you started the replay from the main menu, "Play From Here" on the replay bar starts a free game from that turn with the same score, cube and dice; the opponent plays at the level of the match being replayed, and the free game is not saved and does not count toward the statistics. If the match has more than one game, the top line of the panel shows the game number and the score at the start of that game; the arrows beside it switch to the previous

and next game, and after a game's last turn → moves to the start of the next game, while before its first turn ← goes to the end of the previous one. "Export..." saves the match record in three formats: readable text (.txt), the .mat format read by Jellyfish and XG, and GNU Backgammon's .sgf format. That way you can also study your match in these programs; their records, in turn, come in with "Import...", section 5.6.



FIGURE 5.12 **Move history.** The history of a **session**: the dice commitment below the summary; rows for an automatic double, turns with their dice, a double, a beaver and a take.

Every position has two IDs in GNU Backgammon format: the 14-character position ID carries where the checkers are, the 12-character match ID the score, the cube, whose turn it is and the dice. "Copy IDs" on the pause card, in the review panel and in the position editor copies both to the clipboard, together with the XGID read by eXtreme Gammon and BGBlitz. "Copy Link" gives the position as a link that starts with "pulvezar://konum"; on the device that opens the link, Checkers and Dice sets up the position in the review panel. Paste the IDs into GNU Backgammon or another program that reads this format to set up the exact same position there, or post them on a forum to ask others. It works the other way too: an ID or an XGID you have copied is set up on the board with "Inspect Position from Clipboard" on the "Matches" card, and you see the Master engine's evaluation and, if the match ID has dice, the best moves in the review panel. The "Share" button (on the pause card, in the review panel and on the end-of-game card) shares the position or the result as an image: a board diagram, the score and explanatory lines, with the IDs and the app's web page

underneath. You can send the image from the share menu by Messages, Mail or AirDrop.

On the Mac, the “Open in Window” button in the header of the “Library,” “Move Analysis” and “Move History” cards opens the card in a separate window; the board stays in the main window. The same windows also open from the Window menu with ⌥⌘L, ⌥⌘A and ⌥⌘H; Esc or the close button closes the window. On iPad, if multiple windows are supported (in Stage Manager or side by side), the same button opens a separate window; no board is set up in a second main window, and “Bring the board here” moves the board to that window.

5.9 Settings

You can reach Settings from the main menu or from the pause card. Changes take effect immediately.

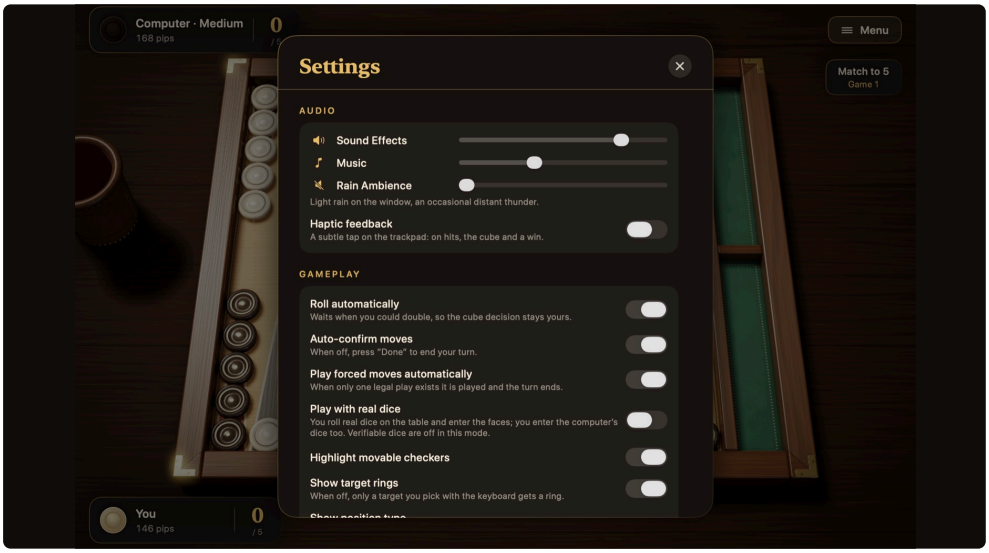


FIGURE 5.13 **Settings.**

- **Audio:** Separate levels for sound effects, music and the sound of rain on the window. “Haptic feedback” gives short taps: on iPad when the dice land, when you hit, when you bear off and on the cube (with a device that supports it or the Magic Keyboard’s trackpad; on by default), and on the Mac on a

Force Touch trackpad when you hit, on the cube and when you win (off by default).

- **Gameplay:** Automatic rolling, auto-confirm and forced moves, playing with real dice, movable checkers, target rings, position type, the Hint and Undo buttons, Coach, warning threshold and natural-language explanations; details in section 5.3. The rules are not in Settings: the two traditional Turkish tavla rules and the match clock are chosen in the New Match panel.
- **Speed:** Animation speed and opponent think time. With "Dice cup" off, the dice are rolled without the cup and rolling is quicker.
- **Appearance:** Board ("Classic Walnut," "Mother-of-Pearl," "Maple," "Ebony Night," "Olive Oak," "Cherry & Honey," "Sandstone"), checkers ("Boxwood & Ebony," "Marble & Onyx," "Red & Black Lacquer"), dice ("Ivory," "Ebony," "Amber") and felt ("Green," "Burgundy," "Navy"). The checker and dice styles change the sound too, and the checker icons on the screen take on the colors of the checker style you choose. "Home board on the left" mirrors the board, and the checkers are borne off to the left. "Point numbers" prints your numbering on the rails.
- **Camera:** Cinematic camera (a push-in at the start of a match, a descending orbit at the end), zoom and camera tilt.
- **Daily Position reminder:** With "Remind me" on, a notification reminding you of the daily position arrives at the time you choose (08:00, 12:00, 18:00, 19:00, 20:00 or 21:00); when you turn it on, the system asks for permission to send notifications. It is off by default.
- **Language and text:** Turkish, English, French, German, Spanish, Portuguese, Italian, Greek and Russian. With "System" selected, the system language is used. Text size can be "System," "Normal," "Large" or "Larger"; "System" follows the system text size setting.
- **iCloud:** With "Sync with iCloud" on, your matches in progress, the archive, the library and your flashcards sync through your iCloud Drive between your devices on the same Apple Account; statistics and achievements carry over too: each device writes its own numbers, and the Statistics card adds them all up. It is off by default. The line below it shows the last sync, and "Sync Now" syncs without waiting. iCloud is only available in the App Store version; if iCloud Drive is off, the line says so.
- **System:** On the Mac, "Show in the menu bar" puts an item in the menu bar: the item shows the score of the match in progress, and its menu has the score, whose turn it is, "Pause" or "Resume," today's status of the daily

position and your streak (choose it to open the position), the Full Rollout queue while one is running, and “Show Window.” On iPad, “Live Activity” shows the match in progress on the Lock Screen: the score, the game and whose turn it is. Both are off by default.

5.10 Keyboard shortcuts

Every move on the board can be played with the keyboard, without using the mouse.

TABLE 5.1 Keyboard shortcuts

Key	What it does
Space	Rolls the dice when a roll is waiting; confirms the turn when auto-confirm is off; in a replay, plays and pauses; in a review, cycles through the position, the best move and the move you played
Return	Moves the checker to the target chosen with the keyboard; confirms the turn when auto-confirm is off
Tab, ⇧Tab	Cycles through the checkers you can move
Arrow keys	Cycle through the selected checker’s targets
Esc	Pauses, and resumes the game when pressed again; closes an open card, the New Match panel and a separate window; leaves a lesson
⌘Z	Undoes the last step; works when the “Undo” button is on
⌘I	Hint; works when the “Hint” button is on
⌘D	Doubles
⌘L	Opens the move history
⌥⌘L, ⌥⌘A, ⌥⌘H	On the Mac, opens the Library, Move Analysis and Move History cards in a separate window
⌘N	New match; asks first while a match is in progress, between games too
⌘?	Opens the “How to Play” card
← →	Previous and next turn in a replay; previous and next mistake in a review

Key	What it does
1–6, ⌫	Enter and delete faces in the dice panel when playing with real dice; Return confirms the entry

5.11 Accessibility

- **Screen reader:** The board can be played with VoiceOver. The points, the bar, the borne-off checkers and the dice are separate elements; for each one, VoiceOver reads out whose checkers are there and how many, and whether it is movable, selected or a target. In a review, in a replay and in the position editor, the board also reads out the position shown at that moment; the small board diagrams on the cards are read point by point too. Your dice, your opponent's moves and hit checkers, the end of an analysis or a simulation, the results of the trainers and the daily position, and the last 30 seconds on the clock are announced aloud; moves are read as "from point 24 to point 18," and scores with each side's name. When a card opens, the focus moves to its title and the elements behind it are hidden; VoiceOver's escape gesture (on iPad, a Z drawn with two fingers) closes the card. With VoiceOver on, pip counting has no time limit.
- **Voice Control:** Every icon button has a name; you can address buttons by the names shown on the screen.
- **Reduced motion:** On the Mac and iPad, if "Reduce motion" is on, camera animations, menu sway and parallax, the dice cup, checker bounces and the gold rain are turned off; cards, panels and notices fade in instead of sliding or growing.
- **Contrast:** On the Mac and iPad, if the "Increase contrast" setting is on, the interface's text and lines are drawn more distinctly. Blunders are marked not only in red but also with a "!!" mark.
- **Text size:** The default "System" option follows the system text size setting (on the Mac in System Settings under Accessibility, Display; on iPad under Display & Brightness); in Settings you can also choose "Large" or "Larger." Cards and indicators grow to match.
- **Keyboard:** Every move on the board can be played with the keyboard; the keys are in Table 5.1.

5.12 Practice: daily position, lessons, flashcards, openings, library, trainers and position editor

The “Practice” card on the main menu brings ten practice tools together: “Daily Position,” “Daily Cube Decision,” “This week’s plan,” “Lessons,” “Flashcards,” “Opening Library,” “Library,” “Score awareness,” “Trainers” and “Position Editor.”

Daily Position. A new position arrives every day; on a given day everyone solves the same position. The position comes separately for each rule set: the tile opens with the rules chosen in New Match and switches with the “Turkish tavla” and “International” chips; the Turkish tavla position is played with the basic rules, with “No hit-and-run” and “Must bear off” off. The dice are given, and you find the best move. The tile shows the match length, the score and the move number; “Play” sets the position up on the board, and you play your move. When the turn ends, a result card appears: if your move was the best, it says “Right move!”, otherwise “There was a better move” or “Blunder”; below are a diagram of the position with the arrows of the best move, the move you played and the best move, and a sentence explaining why. “Show on Board” opens the candidates in the review panel, “Try Again” asks the same position once more, and “Finish” returns to the main menu. The “History” button on the tile opens a monthly calendar: the days you solved, missed and did not attempt are marked; tap or click a day and that day’s position opens on the board, and a past day is always a retry. If the “Daily Position reminder” in Settings is on, a notification arrives at the time you choose (08:00, 12:00, 18:00, 19:00, 20:00 or 21:00); it doesn’t come if you have already attempted the position that day. In each rule set only the first attempt of the day counts, and each rule set has its own streak: if you find the right move, that rule set’s streak grows by a day, and if not, it resets; retries do not count toward the streak or the statistics. For the selected rule set, the Practice card shows whether you have solved today’s position and your streak, and, if you missed it, your loss against the best move; the number of positions solved, the streak, the best streak and the average loss are on the Statistics card, separately for each rule set. The Daily Position widget shows today’s position, your streak, your result and the daily cube decision on the Home Screen or on the desktop; tap or click it and the position opens with that rule set. The small size shows only the streak and the

result, the medium size the position diagram as well; the new position arrives at midnight, with no need to open the app.

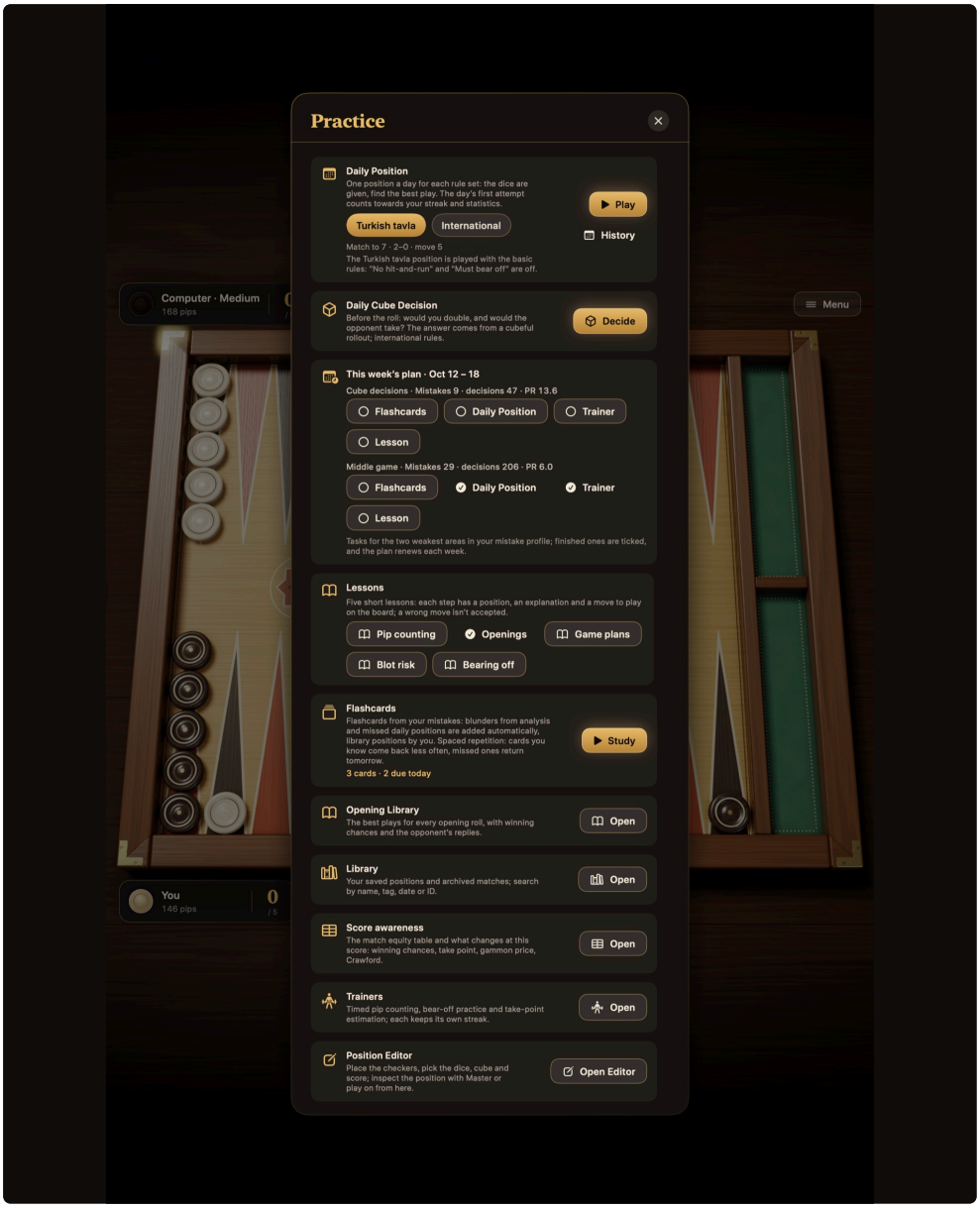


FIGURE 5.14 **Practice card: the daily position tile (rule chips, match length, score, move number and history), the daily cube decision, this week's plan, the lessons, the flashcards (deck summary), the opening library, the library, score awareness, the trainers and the position editor.**

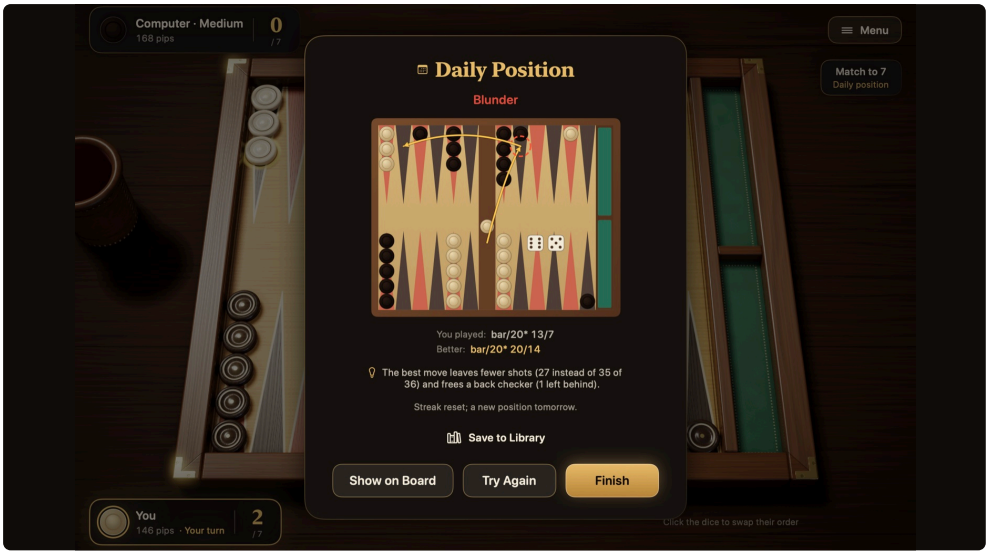


FIGURE 5.15 **Result card of the daily position.** The move played is behind the best: the arrows in the diagram show the best move, the lines below it the move played and the best move, and the sentence with the bulb icon the reason.

Daily Cube Decision. Every day a cube decision arrives, under international rules. You see the position before the roll and the match situation, and you answer the question "Would you double?" and, if you double, "Would the opponent take?" The answer comes from a simulation played out to the end of the game with the cube in play: the result section shows your decision, the simulation's verdict, the equities of the three options and, if you were wrong, your loss. As with the daily position, only the first attempt counts toward the streak and the statistics.

This week's plan. A personal weekly plan is built from the most frequent mistakes in your analyses: the two areas where you lose the most points (at least 40 analyzed decisions per area) and tasks for each: flashcards of that type, the daily position, a suitable trainer and, if there is one, a related lesson. Tasks you finish are checked off, and the plan is renewed every Monday. If there isn't enough data, the card says why it is empty.

Lessons. There are five interactive lessons: "Pip counting," "Openings," "Game plans," "Blot risk" and "Bearing off." A lesson moves forward step by step in the panel beside the board: in reading steps you read the explanation and press "Continue," and in move steps you play the move on the board. A wrong move

isn't accepted, and "Try Again" sets up the same step again; finished lessons are checked off. "Leave lesson" (Esc) exits the lesson.

Flashcards. A deck builds up from your mistakes: blunders from the end-of-game analysis and daily positions you missed become flashcards automatically; "Add to Flashcards" on the analysis card adds your other checker-play mistakes, and the flashcard button in the row of a library position that has dice adds that position. Each flashcard is a question: the position and the dice are given, and you find the best move. The tile shows how many flashcards are in the deck, how many are due today or when the next one comes up, and how many are mastered. "Study" asks the flashcards that are due (at most twenty) one after another; if none are due, it asks the ten due soonest ahead of time. You play a flashcard just like the daily position; the result card shows which flashcard you are on, how many you've gotten right and when this one will be asked again, and "Next Card" moves on to the next one. Spaced repetition: a flashcard you get right comes back after one day, then after six days, then at ever longer intervals; a flashcard you miss comes back the next day. A flashcard whose interval exceeds three weeks counts as mastered. Flashcard study does not count toward the daily position statistics or streak.



FIGURE 5.16 **Result card of flashcard study: at the top, which flashcard this is and how many are right; below, the day the flashcard will be asked again.**

Opening Library. For each of the twenty-one opening rolls it lists the best moves, with their loss against the best and their winning and gammon chances.

The values are the average of Master's 3-ply search and a simulation of 5184 trials played out to the end of the game; the chances come from the simulation. The chips at the top choose the rule set: under international rules there is no opening with doubles, under Turkish tavla there is. Selecting a roll shows the best move in the diagram; "Opponent's replies" opens the best reply to each of the opponent's rolls after that move. "Set Up on Board" sets the opening position up with that roll in review mode: the candidates are animated on the board, and this is how you practice the opening moves of section 4.3. The "You" row below the candidates shows, for the games in your archive that you opened with this roll, the move you played most often, how many times you played it, your win rate and how far it falls short of the library's best. "Quiz," next to the rule chips, gives you a random opening roll: you find the best move on the board, and the result card shows the number of correct answers and your streak.

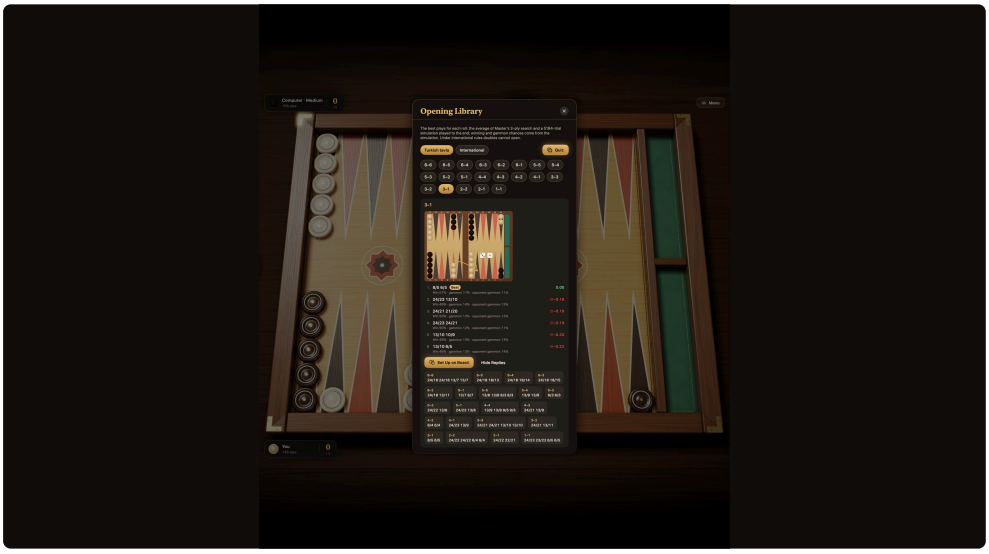


FIGURE 5.17 **Opening Library: rule set and the "Quiz" button, the 21 opening rolls, the diagram of the best move for 3-1, the candidates with winning and gammon chances, and the opponent's replies below.**

Library. The positions you have saved and the matches in the archive are gathered on a single card. You add a position with "Save to Library" on the pause card, in the review panel, in the position editor or on the daily position's result card; the same position is never added twice, and the button changes to "In the library." Each row of the "Positions" tab shows the position's name, its date, where it came from, the match and dice details, its tags and its note: with the pencil you change the name, the tags (comma-separated) and the note, and with

the trash can you delete the position; “Open on Board” sets the position up in the review panel, where the Master engine evaluates it; the button next to it opens “Similar Positions” (section 5.7). The “Matches” tab lists the archive: it separates matches won and lost, shows the approximate PR value of analyzed matches, and “Replay” lets you watch a match again. The search box looks through names, tags, notes, dates and IDs; the chips separate Turkish tavla from international rules. On the Mac you can open the card in a separate window with “Open in Window” (section 5.8); Spotlight also finds the positions in the library.

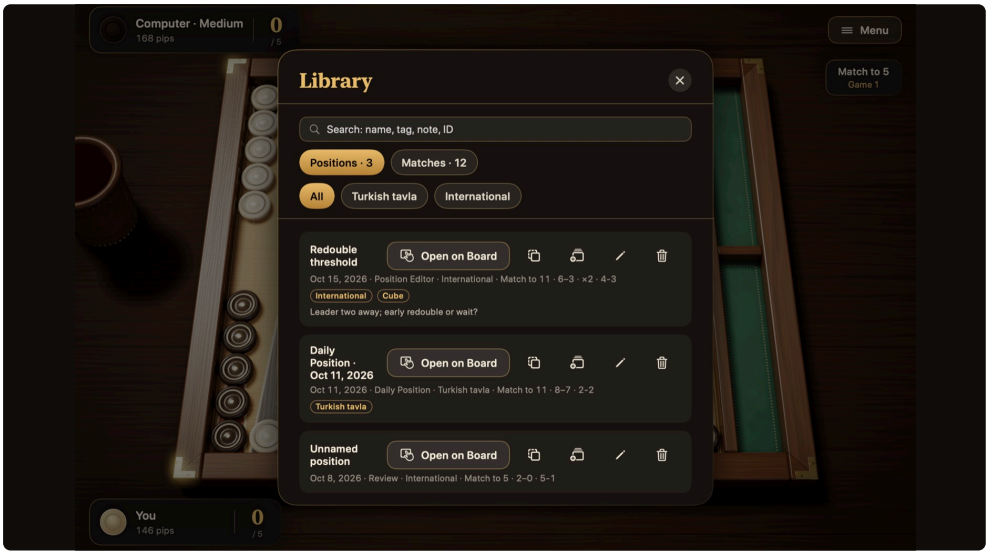


FIGURE 5.18 **Library: search box, positions and matches tabs, rule chips; for each position, the Open on Board, Similar Positions, flashcard, edit and delete buttons beside its name, and below it the date, source, match details, tags and note.**

Score awareness. The tile opens the “Match Equity” card described in section 5.4.

Trainers. There are three short drills. “Pip counting” asks you to enter both pip counts of a position within 20 seconds; with VoiceOver on there is no time limit. The “Bear-off trainer” opens a bear-off position with no contact on the board and judges your move by the exact winning chances from the bear-off database. “Take point” gives you a score and a cube, and you estimate the gammonless take point as a percentage; an answer within 3 percentage points

counts as correct. Each trainer keeps its own counts of correct answers, attempts and streak.

Position Editor. You set up any position you like by hand; you reach the editor from the "Practice" card or with "Open Editor" on the "Matches" card. Pick the "Place light checker," "Place dark checker" or "Remove" tool and click the points and the bar on the board (on an iPad you tap); on a Mac a right click removes a checker. Each side has at most 15 checkers, checkers not on the board count as borne off, and no checker can be placed on a point held by the opponent. Below, you choose whose turn it is, the dice ("No dice" is possible too), the cube's value and owner, the match length, the score and whether the Crawford game has been played. "Start position" sets up the opening arrangement, "Clear" empties the board, "Swap sides" sets the position up from the opponent's point of view, and "Paste" reads a position ID or an XGID from the clipboard; "Copy IDs" and "Copy Link" take the position outside (section 5.8). "Inspect" opens the position in the review panel: you see the Master engine's evaluation and, if you gave dice, the best moves. "Play" starts a free game from the position; a free game is not saved and does not enter the statistics or the archive. For a position that cannot be set up, both buttons are disabled and the reason is shown: when one side has borne off all its checkers, or when the score has reached the match length.

CHAPTER SUMMARY

- For a new match you choose the difficulty (or the “Adaptive” opponent), the opponent style, the rules, an optional match clock, the match length or a session, and your checker color; these choices apply for the whole match.
- The dice are rolled for you, or you can play with real dice; you move checkers by clicking, dragging, double- or right-clicking, or with the keyboard, and your turn ends with your last step.
- The “Hint” and “Undo” buttons are turned on in Settings. “Hint” shows the move the Master engine suggests; under international rules it also gives advice on cube decisions. With “Coach” on, a mistaken move or cube decision is reported before the turn ends, with a chance to undo. The Coach, the analysis and the review say in one sentence why the better move is better; on devices with Apple Intelligence, a language model can make this sentence read more fluently.
- The “Match Equity” card shows your chance of winning the match, the take point and the gammon price at every score.
- Matches are saved automatically; from the “Matches” card you can return to matches in progress and replay finished ones, .mat, .sgf and .xg files can be imported or opened directly, and every game in the archive can be analyzed later. “Analyse All” does the whole archive in one batch. Handoff, the “Continue” widget, Shortcuts and Siri are other ways back to a match.
- The end-of-game card shows the result, the summary, dice luck and the move analysis; mistakes can be reviewed on the board with “Show on Board,” and “Deep Review” re-evaluates the candidates with thousands of simulated games. “Full Rollout” plays them out to the end of the game and, in a match or a session, cube decisions too. “Similar Positions” finds similar positions in the archive, and “Play From Here” starts a free game from the turn being reviewed or replayed. The match record is exported as .txt, .mat or .sgf; the position and the result can be shared as IDs, an XGID, a link or an image, and the dice can be verified once the game is over. Twenty achievements, your rating, your

most frequent mistakes and the PR trend are tracked on the Statistics card.

- Settings is made up of audio, gameplay, speed, appearance, camera, daily position reminder, language, iCloud and system sections: seven boards, nine languages; every move can also be played with the keyboard.
- The “Practice” card brings together the daily position and its history, the daily cube decision, the weekly plan, the lessons, the flashcards, the opening library and its quiz, the library, score awareness, the trainers and the position editor; a free game started from the editor or with “Play From Here” is not saved.



CHAPTER 6

Reference

This chapter is for reference: frequently asked questions, a glossary and an index. The "See" column of the glossary takes you to the sections where each term is explained.

6.1 Frequently asked questions

I rolled badly. Can I pass? No. If the dice allow a move, you must make it. A pass happens on its own, and only when no die can be played.

Can I move my opponent's checkers? No. Each player moves only their own checkers.

After hitting a checker, can I carry on with the same checker? Under the international rule, yes: hitting doesn't end the checker's journey, and you can move it on with the remaining die. In the common form of Turkish tavla, however, hit-and-run in your own home board is forbidden. In Checkers and Dice, when you play by Turkish rules, this restriction applies if "No hit-and-run" was selected for the new match; it is off by default.

What is the most checkers a point can hold? There is no limit. You can stack as many checkers as you like on a point of your own.

Can I start bearing off while I have a checker on the bar? No. To bear off, all your checkers must be in your home board; a checker on the bar counts as outside it.

Do I have to add the two dice together and play them with one checker? No. You can also give the two dice to two different checkers. If you play them with one checker, the intermediate point must be open.

Can a game end in a draw? No. Because players take turns, one of them always finishes first.

I had to break a point in my own home board. Was that a mistake? Not always. Sometimes the dice leave you no other option; the forced-move rule comes before everything else.

I moved a checker by mistake in Checkers and Dice. What should I do? With the default settings, a step you've played can't be taken back. To be able to undo, turn on "Show the Undo button" in Settings; you can then take back that turn's steps until the turn ends. While "Auto-confirm moves" is on, the turn ends with your last step; if you turn that off too, you can undo until you press "Done."

Why did my pip count suddenly go up? One of your checkers has been hit. A checker sent to the bar goes back to the start of the route, so your pip count rises by the distance that checker has lost.

Are the dice rigged? No. The dice are rolled on this device: at the start of every game a secret seed is chosen with the operating system's random number generator, and all the dice of both sides are derived from that seed; nothing is adjusted by difficulty level. While the game is in progress, Move History shows the seed's commitment, and once the game is over, the seed itself; "Verify" checks that the dice in the record came from that seed (section 5.8). In short runs, back-to-back doubles or bad rolls are normal; over time the distribution evens out. The Statistics card shows how often each face came up, side by side with the expected count per face.

Can I review a match I played in another program in Checkers and Dice? Yes. You bring a Jellyfish or XG .mat file, an XG .xg file or a GNU Backgammon .sgf match record into the archive with "Import..." on the "Matches" card or by opening the file directly; you replay its games and have the Master engine analyze them with "Analyse." Details in section 5.6.

Can I set up a position myself and ask about it? Yes. With the "Position Editor" on the "Practice" card you set up the checkers, the dice, the cube and the score, and with "Inspect" you see the Master engine's evaluation and the best moves; details in section 5.12.

Do my matches carry over between my Mac and my iPad? If you want them to. With "Sync with iCloud" on in Settings, your matches in progress, the archive, the library and your flashcards sync through your iCloud Drive between your devices on the same Apple Account; statistics and achievements carry over too and are shown as the total across your devices. Syncing is off by default and is only available in the App Store version. You can also carry on a match in progress on your other device with Handoff; that needs no iCloud sync (section 5.6).

6.2 Glossary and index

TABLE 6.1 Glossary and index

Term	Meaning	See
Achievement	One of twenty rewards earned by reaching a goal in the game, shown as a badge on the Statistics card	5.7
Adaptive opponent	The fifth opponent, which picks its strength from the game grade of your recent games or from your rating; the strength stays the same for the whole match	5.1
Anchor	A made point of two or more checkers held in the opponent's home board	4.4
Backgammon	A gammon in which, on top of that, the loser still has a checker on the bar or in the winner's home board; 3 points	3.7
Bar	The raised strip that divides the board in two; hit checkers wait here	2.1, 3.5
Batch analysis	Analysis of every game in the archive not yet analyzed, one after another, in the background	5.6
Bearing off	Taking the checkers in the home board off the board	3.6, 4.6
Beaver	When a player who has been doubled immediately doubles the cube once more	3.8, 5.4
Blot	A checker standing alone on a point, which can be hit	3.4, 4.5

Term	Meaning	See
Board theme	The look of the board; seven options in the Appearance section of Settings	5.9
Checker	The round pieces the players move; each player has 15	1.5
Checker on the bar	A checker that has been hit and placed on the bar, waiting to re-enter the game	3.5
Closed out	When all six points of the opponent's home board are made; a checker on the bar can't enter	3.5
Coach	A learning mode that reports a mistaken move or cube decision before the turn ends and lets you undo	5.3
Crawford game	The first game played after one of the players first comes within 1 point of the target; the cube is not used	3.8, 5.4
Daily position	A position that comes to everyone each day, separately for each rule set: the dice are given and the best move is sought; in each rule set the day's first attempt counts toward that rule set's streak and statistics	5.12
Dance	Failing to enter a checker from the bar with either die	3.5
Deep review	Re-evaluation of the candidates in the review panel with thousands of simulated games; the noise margin is shown beside the result	5.7
Dice commitment	A hash of the secret dice seed fixed at the start of the game; when the game ends, the seed is revealed and the dice are verified	5.8
Dice luck	The difference, in points, between the value of a roll's best move and the average over the 21 possible rolls	5.7
Double	An offer to double the value of the game	3.8, 5.4
Doubles	Both dice showing the same number; four moves are played	3.2
Flashcard	A "find the best move" question made from your mistakes and asked with spaced repetition	5.12
Forced move	The only move that can be made under the rules	3.3
Free game	A game started from the position editor or with "Play From Here"; it is not saved and does not enter the statistics	5.12
Full rollout	Re-evaluation of the candidates or, in a match, of the two branches of a cube decision, by playing them out to the end of the game, with variance reduction	5.7

Term	Meaning	See
Gammon	Finishing the game before the opponent has borne off any checkers; 2 points	3.7
Hit	Landing a checker on an opposing blot and sending it to the bar	3.4
Hit-and-run	Carrying on with the same checker after hitting a blot	3.4
Home board (inner board)	Points 1–6, where the checkers are borne off	2.1, 3.6
iCloud sync	Carrying matches in progress, the archive, the library, flashcards, statistics and achievements between devices on the same Apple Account through iCloud Drive; optional	5.9
Importing	Bringing a Jellyfish, XG (.mat, .xg) or GNU Backgammon (.sgf) match record into the archive	5.6
Jacoby rule	Gammons and backgammons count as a single game while the cube is in the middle (no double has been taken); found in sessions, not used in matches	3.8, 5.1
Library	A searchable list of saved positions and archived matches	5.12
Made point	A point held by at least two checkers of the same color, where the opponent can't land	3.1, 3.4
Match clock	Time per side and a delay per move; a side whose time runs out loses the game at the cube value	5.1
Match equity	The chance of winning the match at a given score; the take point and the gammon price follow from it	5.4
Move explanation	The sentence in the analysis, the review and the Coach card that says why the best move is better; optionally made more fluent by the on-device language model	5.3
Moving instead of bearing off ("sallamak")	Choosing to move checkers within the home board when a checker could be borne off	3.6
Opening library	Practice tool giving the best moves, their chances and the opponent's replies for the twenty-one opening rolls	4.3, 5.12
Opponent style	The computer's leaning among moves that are close to each other: balanced, aggressive, cautious or runner	5.1

Term	Meaning	See
Outer board	Points 7–12, just in front of the home board	2.1
Pip	The total number of points a player must move to bear off all of their checkers	4.2
Ply	How many turns ahead a search, or the choice of moves in a simulation, looks, counting each player's turn as one; 1-ply is direct evaluation	5.7
Point	Each of the 24 triangles on the board	2.1
Position editor	Tool for setting up the checkers, dice, cube and score by hand, to inspect the position with Master or play on from it	5.12
Position ID, match ID	Short codes that write the position and the match state in GNU Backgammon format (the XGID carries the same information in eXtreme Gammon format); they set up the same position in other programs	5.6, 5.8
PR	An approximate error rate the Master engine computes from the loss in your decisions; lower is better	5.7
PR trend	The approximate PR of the analyzed matches in the archive, drawn as a line over time	5.7
Prime	Made points lined up side by side; they block the way for the opponent's back checkers	4.4
Raccoon	When the player facing a beaver immediately doubles the value once more	3.8, 5.4
Rating	A local Elo-style score calculated from games against the computer; it starts at 1500, and the tiers run from Apprentice to Grandmaster	5.7
Spaced repetition	Bringing back a flashcard you got right at ever longer intervals, and one you missed the next day	5.12
Variance reduction	Shrinking the noise in a simulation by subtracting each roll's luck from the result	5.7

6.3 Afterword

If you've come this far, you now know all the rules of backgammon, the basic plans and every corner of Checkers and Dice. The rest is learned at the board. Play your first matches on "Easy"; look over the analysis of the games you lose to see which decisions cost you points, then raise the level. The dice smile on you sometimes and on your opponent at other times; in the long run, your choices make the difference.

For five thousand years, backgammon has been bringing people together on either side of a table. I hope Checkers and Dice will be a small contribution to this tradition and bring you many enjoyable hours. Your feedback and suggestions help make the app better.

Enjoy your games!